

Microhomology-Mediated Circular DNA Formation from Oligonucleosomal Fragments During Spermatogenesis

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Abstract

The landscape of extrachromosomal circular DNA (eccDNA) during mammalian spermatogenesis, as well as the biogenesis mechanism remains to be explored. Here, we revealed widespread eccDNA formation in human sperms and mouse spermatogenesis. We noted that germline eccDNAs are derived from oligonucleosomal DNA fragmentation in cells likely undergoing cell death, providing a potential new way for quality assessment of human sperms. Interestingly, small-sized eccDNAs are associated with euchromatin, while large-sized ones are preferentially generated from heterochromatin. By comparing sperm eccDNAs with meiotic recombination hotspots and structural variations, we found that they are barely associated with *de novo* germline deletions. We further developed a bioinformatics pipeline to achieve nucleotide-resolution eccDNA detection even with the presence of microhomologous sequences that interfere with precise break-point identification. Empowered by our method, we provided strong evidence to show that microhomology-mediated end joining is the major eccDNA biogenesis mechanism. Together, our results shed lights on eccDNA biogenesis mechanism in mammalian germline cells.

eLife assessment

This study provides **important** information on the biogenesis of eccDNAs during spermatogenesis. The data presented are **solid** and supportive of the conclusion that eccDNAs in spermatogenic cells are not derived from mitotic recombination hotspots but rather represent oligonucleosomal DNA fragments from apoptotic male germ cells, whose ends are ligated through microhomology-mediated end-joining. This work is of interest to researchers working on germ cell biology and cancer biology.

Introduction

Apart from linear chromosome, extrachromosomal DNA in circular form also exists in the nuclei of eukaryotes (Noer *et al.*, 2022 [↗](#)). Circular DNAs could be roughly classified into two groups based on their cell origins (Chiu *et al.*, 2020 [↗](#)). The one exclusively present in cancerous cells is usually referred to as ecDNA (extrachromosomal DNA), which is of megabase long in average and plays key roles in tumorigenesis (T. Wang *et al.*, 2021 [↗](#)). The ecDNA biogenesis is linked to ‘episome model’ (Carroll *et al.*, 1988 [↗](#)), chromothripsis (Shoshani *et al.*, 2021 [↗](#)), breakage-fusion-bridge (Coquelle *et al.*, 2002 [↗](#)), or translocation-deletion-amplification model (Van Roy *et al.*, 2006 [↗](#)). The other class found in somatic and germ cells is usually called eccDNA (extrachromosomal circular DNA). The size of eccDNA ranges from dozens of bases to hundreds of kilobases (Paulsen *et al.*, 2018 [↗](#)). Contrary to ecDNAs, the biogenesis mechanisms and biological functions of eccDNAs are relatively less experimentally characterized, and current studies show inconclusive or even contradictory results.

The genomic origins of eccDNAs have been extensively investigated in different cells and conditions with the application of Circle-seq and its refined derivatives (Mehta *et al.*, 2020 [↗](#); Moller *et al.*, 2015 [↗](#); Y. Wang *et al.*, 2021 [↗](#)), where eccDNAs are detected via rolling circle amplification and deep sequencing. While CpG islands (Shibata *et al.*, 2012 [↗](#)), gene-rich regions (Moller *et al.*, 2018 [↗](#)), and repeat elements, *e.g.*, LTR (long terminal repeat) (Moller *et al.*, 2015 [↗](#)), LINE-1 (Dillon *et al.*, 2015 [↗](#)), segmental duplication (Mouakkad-Montoya *et al.*, 2021 [↗](#)), or satellite DNA (Mouakkad-Montoya *et al.*, 2021 [↗](#)), are hotspots for eccDNA formation, others found that eccDNAs are nearly random with regard to genomic distribution (Moller *et al.*, 2020 [↗](#)), or even made opposite observations (Henriksen *et al.*, 2022 [↗](#)). Epigenomically, the overall higher GC content, the periodicities of dinucleotide and eccDNA size convergently point to that nucleosome wrapping of DNA might contribute to the formation of small-sized eccDNAs (Shibata *et al.*, 2012 [↗](#); Y. Wang *et al.*, 2021 [↗](#)), an intriguing starting point for mechanistic understanding of eccDNA origination. However, direct evidence for coincident positioning of eccDNAs and nucleosomes is still lacking, not to mention specific epigenetic marks on nucleosomes that are tightly associated with eccDNA formation.

EccDNAs are increased upon DNA damages (Moller *et al.*, 2015 [↗](#); Paulsen *et al.*, 2021 [↗](#)), suggesting them as byproducts of successive DNA repairs. Among diverse repair pathways, it was reported that eccDNA levels particularly depend on resection after double-strand DNA break (DSB) and repair by microhomology mediated end joining (MMEJ) (Paulsen *et al.*, 2021 [↗](#); Y. Wang *et al.*, 2021 [↗](#)). In further support of the involvement of MMEJ, microhomology is found around eccDNA break-points (Lukaszewicz *et al.*, 2021 [↗](#); Moller *et al.*, 2015 [↗](#)). However, in all studies using short-read sequencing technologies, eccDNA break-points are mis-annotated if microhomologous sequences are present around due to their interference to precise break-point detection (see our main text) (Dillon *et al.*, 2015 [↗](#); Henriksen *et al.*, 2022 [↗](#); Kumar *et al.*, 2017 [↗](#); Lv *et al.*, 2022 [↗](#); Mann

et al., 2022 [\[1\]](#); *Moller et al., 2018* [\[2\]](#); *Moller et al., 2020* [\[3\]](#); *Paulsen et al., 2019* [\[4\]](#); *Prada-Luengo et al., 2019* [\[5\]](#); *Shibata et al., 2012* [\[6\]](#); *Sin et al., 2020* [\[7\]](#); *Y. Wang et al., 2021* [\[8\]](#); *P. Zhang et al., 2021* [\[9\]](#)), or the eccDNA identification doesn't depend on precise break-point detections at all (*Moller et al., 2015* [\[10\]](#); *Mouakkad-Montoya et al., 2021* [\[11\]](#)). The contribution of microhomology to eccDNA generations thus needs to be revisited with precise mapping of break-points.

Alternative or additional mechanisms might be involved in germline eccDNA formation. During meiosis, two spatially-closed break sites catalyzed by SPO11 at recombination hotspots may release eccDNAs accompanied by *de novo* deletions on linear chromosomes (*Lukaszewicz et al., 2021* [\[12\]](#)). Consistently, germline microdeletions display similar sequence features with eccDNAs (*Shibata et al., 2012* [\[6\]](#)). However, a recent study reported that the creation of germline eccDNAs negatively correlate with meiotic recombination rates (*Henriksen et al., 2022* [\[13\]](#)). Therefore, it remains to be determined whether meiosis might significantly contribute to eccDNA biogenesis.

We envision that eccDNA landscape during spermatogenesis is ideal for clarifying the abovementioned issues and so better understand the biogenesis mechanisms and biological implications of eccDNAs. Only a small fraction of histones will survive from the histone-to-protamine transition in mature sperms, allowing us to more specifically correlate eccDNA origination with histones. Studying eccDNAs in germline cells rather than somatic cells could help reveal to what extent meiosis might contribute to eccDNA generation and *de novo* structural variations that can be passed to offspring. Therefore, in this study, we profiled eccDNAs via Circle-seq in human sperms and different developmental stages of mouse germ cells with an improved analysis pipeline to identify eccDNAs at nucleotide resolution. We conclude that germline eccDNAs are likely formed by microhomology mediated ligation of nucleosome-protected fragments, and barely contribute to *de novo* genomic deletions at meiotic recombination hotspots.

Results

Widespread eccDNA formation in human and mouse germline cells

Because it was reported that high content of eccDNAs existed in sperms (*Henriksen et al., 2022* [\[13\]](#)), we examined the genome-wide eccDNA landscape in two human sperm samples by Circle-seq (*Moller et al., 2018* [\[2\]](#); *Moller et al., 2015* [\[10\]](#); *Moller et al., 2020* [\[3\]](#)) (see Materials and methods), and indeed found that there were widespread eccDNAs across the human genome (**Figure 1A** [\[14\]](#), **Figure 1—figure supplement 1A** [\[15\]](#)). This motivated us to further investigate the biogenesis mechanism, particularly whether it might be linked to specific spermatogenesis processes. Given that it is ethically prohibited and technically challenging to collect pure spermatogenic cell types from human individuals, we therefore turned to use mouse model to study the eccDNA formation during spermatogenesis.

A series of cell divisions and morphological changes are involved in spermatogenesis, where spermatogonial stem cells develop into spermatocytes (SPA) via mitosis, and SPA then undergo meiosis to produce haploid round spermatids (RST), which will take a dramatic morphological change and chromatin compaction to produce elongated spermatids (EST) and finally matured sperms (*Hess et al., 2008* [\[16\]](#); *Roosen-Runge, 1962* [\[17\]](#)). We isolated SPA, RST and EST using flow cytometry and collected sperms from mouse cauda epididymis (*Hayama et al., 2016* [\[18\]](#)) for subsequent Circle-seq (*Moller et al., 2018* [\[2\]](#); *Moller et al., 2015* [\[10\]](#); *Moller et al., 2020* [\[3\]](#)) (see Materials and methods). All four cell types were validated with known markers and cell morphology (**Figure 1A** [\[14\]](#)). EccDNA isolation procedures were validated by a high ratio of an exogenous circular DNA (pUC19) to a linear DNA locus (H19 gene) (**Figure 1—figure supplement 2A** [\[19\]](#)), and the low abundance of mitochondria DNA that was supposed to be cleaved by *PacI* and degraded by exonuclease (**Figure 1—figure supplement 2B** [\[20\]](#)). To account for sample variations, up to five biological replicates and ~150 million reads for each cell type were sequenced for

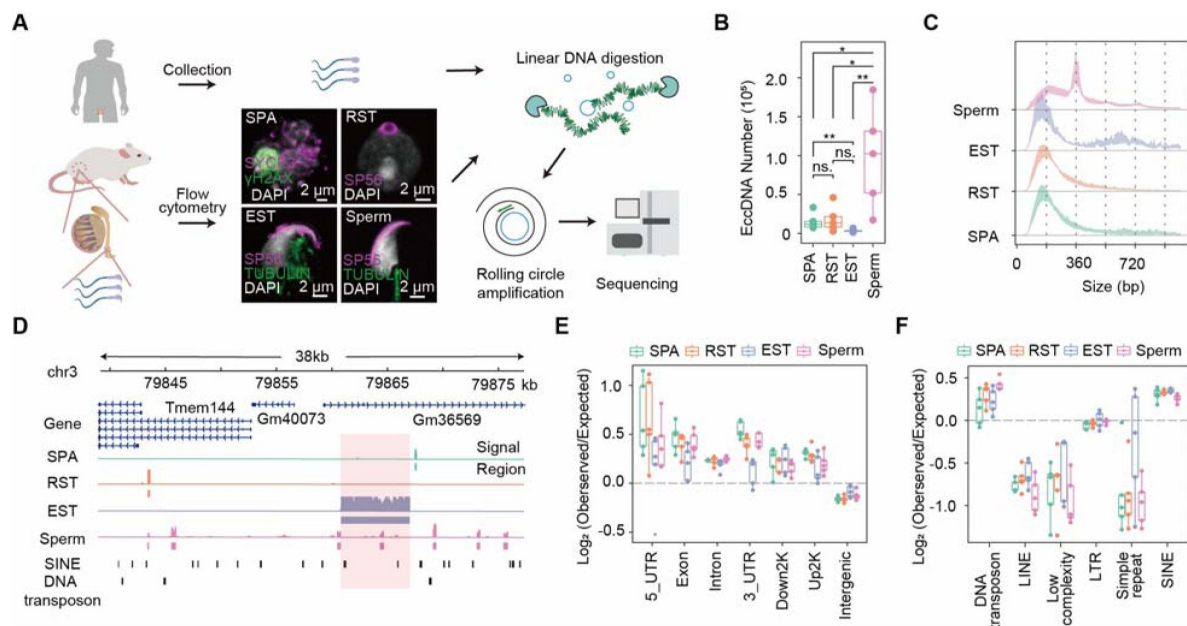


Figure 1.

Overview of eccDNA formation during mouse spermatogenesis. **(A)** Schematic representation of Circle-seq in human sperm cells and mouse SPA, RST, EST and sperm cells validated with immunochemistry. SYCP3: a component of the synaptonemal complex; γH2AX: a marker for double-strand breaks; SP65: a marker for acrosome organelle; TUBULIN: structural component of manchette in EST and flagellum axoneme in sperm cells. **(B)** Number of eccDNAs detected in different cell types. *two-sided *t*-test *p*-value < 0.05; ** two-sided *t*-test *p*-value < 0.01. **(C)** Size distribution of eccDNAs during mouse spermatogenesis. Dotted lines indicate multiples of 180 bp. **(D)** A representative genomic locus showing the gene annotation, Circle-seq signals, detected eccDNAs, and SINE and DNA repeat elements. Highlighted in red rectangle is a large-sized eccDNA. **(E)** Enrichment of eccDNAs at given genomic regions relative to randomly-selected control regions. **(F)** Enrichment of eccDNAs at given repeat elements relative to randomly-selected control regions.

eccDNA detection. From ~1,500 to ~180,000 high-confidence eccDNAs were identified, suggesting widespread circular DNA formation during mouse spermatogenesis (**Figure 1B** [↗](#); see Materials and methods). Some randomly-selected eccDNAs were validated with PCR using outward primers (**Figure 1—figure supplement 2C** [↗](#)). The reproducible rate of eccDNAs with 50% reciprocal overlap between biological replicates was only ~2.4% in average, a level comparable to previous studies (*Henriksen et al., 2022* [↗](#); *Moller et al., 2018* [↗](#)) (**Figure 1—figure supplement 1B** [↗](#)). As noted earlier (*Moller et al., 2018* [↗](#)), the detected eccDNAs seemed not saturated (**Figure 1—figure supplement 1C** [↗](#); see Discussion and Materials and methods), which might underlie the observed low reproducibility. Nevertheless, principal component analysis suggested that the within-group similarity was marginally higher than the between-group similarity (**Figure 1—figure supplement 1D** [↗](#)), allowing investigation of stage-specific eccDNA features during mouse spermatogenesis.

The detected germline eccDNAs verified known genomic features of eccDNAs. First, the natural size distribution of eccDNA is usually distorted in Circle-seq as smaller eccDNAs tend to be overrepresented in rolling circle amplification (*Mohsen et al., 2016* [↗](#); *Moller et al., 2018* [↗](#); *Moller et al., 2015* [↗](#)). As expected, the detected eccDNA population was dominated by small-sized eccDNAs, most of which were ~180bp or ~360bp long (**Figure 1C** [↗](#)). However, eccDNA size could occasionally reach to tens of kilobases (**Figure 1D** [↗](#)). Second, eccDNAs from different cell types were all enriched at gene-rich regions, especially 5'UTR (**Figure 1E** [↗](#), **Figure 1—figure supplement 1E** [↗](#)), corroborating the reported association between eccDNA frequency and gene density in somatic cells (*Dillon et al., 2015* [↗](#); *Shibata et al., 2012* [↗](#)). EccDNAs were also highly associated with SINE but not LINE elements (**Figure 1F** [↗](#)), and quantitative analysis revealed that eccDNA biogenesis was positively correlated with SINE density (**Figure 1—figure supplement 1F** [↗](#)), but negatively correlated with LINE density (**Figure 1—figure supplement 1G** [↗](#)). Given that SINE and LINE elements function to orchestrate chromosomes into gene-rich A compartment and gene-poor B compartment respectively (*Lu et al., 2021* [↗](#)), the positive correlation between eccDNAs and SINE elements might further support that eccDNAs are overall highly associated with the gene-rich regions. Interestingly, we also noticed a strong association between eccDNAs and DNA transposons (**Figure 1F** [↗](#)), suggesting that DNA transposons might get circularized rather than or in addition to re-integrated into the genome, an interesting possibility awaiting further investigations. Altogether, the genome-wide eccDNA landscape during mouse spermatogenesis allow us to further study the biogenesis mechanism and function of eccDNAs.

High eccDNA load and periodic eccDNA size distribution in mouse sperm cells

Notably, sperm cells had 97,372 eccDNAs detected in average, a number significantly higher than those in SPA (15,246), RST (18,426) and EST (3,591) cells (**Figure 1B** [↗](#)). SPA cells did not show higher eccDNA numbers (**Figure 1B** [↗](#)), suggesting that meiosis does not seem to contribute significantly to eccDNA biogenesis. Since the same amount of eccDNAs (10ng) were used for library construction and all samples were sequenced in comparable and sufficiently-deep depth, it suggests that eccDNA species in sperm cells has higher complexity. However, the higher starting cell number for sperm cells might account for the larger diversity of sperm eccDNA species (see Discussion and Materials and methods), otherwise, it would be interesting to explore any specific features of sperm cells underlying the higher load of eccDNAs.

In contrast to SPA, EST and RST eccDNAs showing the unimodal distribution that was centered at ~180bp, sperm-derived eccDNAs showed a multimodal distribution with a pronounced periodicity of ~180bp (**Figure 1C** [↗](#)), which was readily seen in individual samples (**Figure 1—figure supplement 3** [↗](#)). Given that each nucleosome consists of 147bp DNA wrapping itself around a histone core, the ~180bp-long fragments likely corresponded to histone core region plus ~20-30bp linker regions, as observed in apoptotic cells (*Matassov et al., 2004* [↗](#)). Although the identified

eccDNAs in all spermatogenesis stages were likely related to nucleosomes, the different modes of size distribution might be due to distinct nucleosome compositions and structures between sperm and other spermatogenic cells.

Mouse sperm eccDNAs come from DNA fragments protected by histones

Only a small fraction of histones will be retained in mouse sperm cells after histone-to-protamine transition ([Torres-Flores et al., 2020](#)), permitting us to more specifically correlate eccDNAs with histones. We were therefore motivated to see whether the detected eccDNAs were derived from the retained histones in mature sperm cells. We noted that sperm eccDNAs had higher GC content than surrounding regions as well as control regions randomly selected across the genome (**Figure 2A**), resembling the sequence feature of nucleosome-protected DNA fragments. Consistently, sequence-based prediction revealed significantly higher nucleosome occupancy probability for ~180 bp (from 175bp to 185bp) and ~360bp (from 355bp to 365bp) sperm eccDNA regions (**Figure 2B**; see Materials and methods). A small dip was observed at the center of ~360bp eccDNA regions, which likely corresponded to the linker region between two nucleosomes (**Figure 2B**, right).

It is a common practice to re-use publicly-available genomics data generated in the same cell types for integrative analysis. Taking advantage of public ChIP-seq data for histones and their modifications in mouse sperm cells ([Jung et al., 2019](#); [Jung et al., 2017](#); [Singh et al., 2021](#)), we found that eccDNAs were significantly enriched with certain histone variants and modifications (**Figure 2C**), and 7.46% of sperm eccDNAs in total were intersected with at least one ChIP-seq peaks. Considering that histones occasionally retained in sperms might not generate strong ChIP-seq signals exceeding the peak calling cutoff, a meta-gene analysis of ChIP-seq signals at and around sperm eccDNA regions will likely provide more insights. Interestingly, enrichment of H3 histone and H2A.Z, TH2A and TH2B histone variants but depletion of H3.3 variant was observed at ~180bp sperm eccDNA regions (**Figure 2D**). These eccDNAs also showed strong associations with H3K27ac, H3K4me1, H3K9ac and H3K27me3 modifications, however, no enrichment was seen for H3K4me3, and H3K36me3 and H3K9me3 signals were comparable with or even lower than randomly-selected regions as control (**Figure 2D**).

We next examined ~360bp sperm eccDNAs which supposedly correspond to two nucleosomes, and made similar observations. Centers of ~360bp eccDNAs were well positioned between two adjacent nucleosomes consisting H3 histone and H2A.Z histone variants, and H3K27ac, H3K4me1, H3K9ac and H3K27me3 histone modifications (**Figure 2E**). Similar to ~180bp eccDNAs, ~360bp eccDNAs did not show association with H3.3 or H3K4me3, or stronger association than randomly-selected regions with H3K36me3 and H3K9me3 either (**Figure 2E**). Although H3.3 variant coincides with active transcription, it is also well known for its localization at heterochromatin region and its roles in promoting heterochromatin formation by inhibiting H3K9/K36 histone demethylase ([Udugama et al., 2022](#)). Together, euchromatin is generally more preferred than heterochromatin for eccDNA biogenesis, which is consistent with the enrichment of sperm eccDNAs at gene-rich regions (**Figure 1D**).

Large-sized eccDNAs are preferentially generated from heterochromatin regions

Intriguingly, periodic distribution of nucleosomes, *e.g.*, those marked with H3K27me3, was observed for ~360bp but not for ~180bp eccDNAs, indicating that eccDNAs from di-nucleosomes but not mono-nucleosomes preferentially originate from well-positioned nucleosome arrays (**Figure 2E**). We were further prompted to ask whether eccDNAs of different sizes are originated from different genomic regions. Indeed, small-sized eccDNAs (<3kb) were more enriched at H3K27ac-marked euchromatin regions, while large-sized ones (≥3kb) at H3K9me3-marked

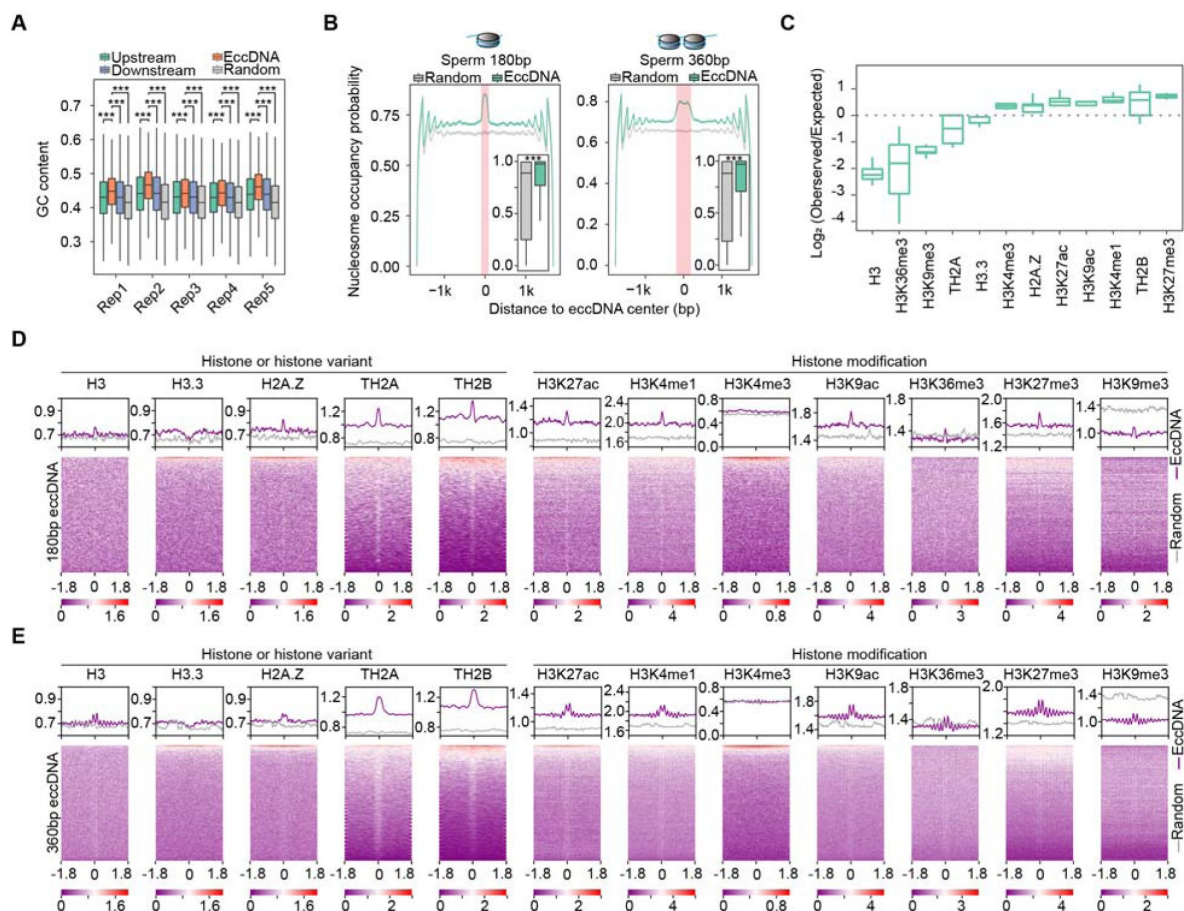


Figure 2.

Association between sperm eccDNAs and nucleosome positioning. **(A)** GC contents of sperm eccDNAs, regions upstream and downstream of eccDNAs, and randomly-selected length-matched control regions. ***two-sided *Wilcoxon* test p -value <0.001. **(B)** Predicted probability of nucleosome occupancy for eccDNA and randomly-selected length-matched control regions (highlighted by red-shaded area), and surrounding regions. Boxplots showing the probability distribution of individual eccDNAs and control regions. ***two-sided *Wilcoxon* test p -value <0.001. **(C)** Enrichment of eccDNAs at different histones and histone modifications. **(D and E)** ChIP-seq signal distribution at [-1.8kb, +1.8kb] of the centers of ~180bp **(D)** and ~360 bp **(E)** eccDNAs. ChIP-seq signals quantified as reads density are color-coded below heatmaps.

heterochromatin regions (**Figure 3A**). Accordingly, small-sized eccDNAs were generally more associated with genic regions, while large-sized ones with non-genic regions (**Figure 3B**). Since sperm eccDNAs in this study were dominantly small-sized ones (**Figure 1C**), strong enrichment of eccDNAs at genic regions was observed (**Figure 1E**). However, strong depletion at genic regions was reported for human sperm eccDNAs in a recent study (Henriksen *et al.*, 2022). Close inspection suggests that the discrepancy is partially reconciled in the light of two eccDNA groups of different sizes. Henriksen *et al.* studied eccDNAs with the size largely ranging from ~3kb to 50kb (Henriksen *et al.*, 2022), rather than small-sized ones reported by us and many others (Dillon *et al.*, 2015; Moller *et al.*, 2018; Moller *et al.*, 2020; Paulsen *et al.*, 2019; Shibata *et al.*, 2012; Y. Wang *et al.*, 2021). This was why we chose 3kb as the cutoff to separate eccDNAs into small- and large-sized categories. In support of this notion, the large-sized sperm eccDNAs detected in this study displayed a weak negative correlation with gene density or Alu elements (**Figure 3C** and **D**). Altogether, compared to euchromatin regions, heterochromatin regions are probably too condensed to be fragmented into smaller pieces for small-sized eccDNA formation.

Germline eccDNAs as apoptotic products are not associated with meiotic recombination hotspots

The observed association between eccDNA and oligonucleosomal DNA fragmentation (**Figure 2**) is a typical feature of cell death. The spontaneous death of germ cells has been observed during the normal spermatogenesis (Liu *et al.*, 2017; Shaha *et al.*, 2010; Weinbauer *et al.*, 2001; Young *et al.*, 2001), however, it is still debatable whether spermatids and sperm can undergo apoptosis (Lachaud *et al.*, 2004). Thus, sperm-derived eccDNAs might be associated with apoptosis (if exists) or un-programed cell death of germ cells during the spermatogenesis (see also Discussion). In support of this hypothesis, all features associated with mouse germline eccDNAs identified in this study (**Figure 1C**, **E** and **F**) closely matched with those of eccDNAs whose generation is dependent on apoptotic DNA fragmentation (**Figure 4—figure supplement 1**) (Y. Wang *et al.*, 2021).

During meiosis, two spatially-closed cleavage sites catalyzed by SPO11 at recombination hotspots could release eccDNAs and generate *de novo* genomic deletions (Lukaszewicz *et al.*, 2021), which, if transmitted to offspring, might contribute to structural variations within population. Since most sperm eccDNAs likely result from oligonucleosomal DNA fragments in sperm cells undergoing cell death (**Figures 2** and **3**) and SPA cells undergoing meiosis does give rise to more eccDNAs than other cells (**Figure 1B**), meiotic recombination is unlikely the major mechanism for germline eccDNA generation. To test this hypothesis, we first investigated to what extent eccDNA break-points well correspond to recombination hotspots defined as SPO11 or PRDM9 binding sites (Alleva *et al.*, 2021; Lange *et al.*, 2016). We noted that there was only a small number of eccDNAs with both break-points located in one recombination hotspot or two different hotspots (**Figure 4A**). These eccDNAs only constituted <0.15% (or <350) of mouse germline eccDNAs, suggesting a very low level of coincidence between eccDNA generation and meiotic recombination (**Figure 4A**). Consistently, only dozens of, or a few hundred eccDNAs in mouse germline cells coincided with known genomic deletions within mouse population (**Figure 4B**). Altogether, germline eccDNAs are likely apoptotic products that are not associated with meiotic recombination hotspots and heritable genomic deletions.

Microhomology directed ligation is the major biogenesis mechanism of germline eccDNAs

We therefore further explored how nucleosome-protected DNA fragments get circularized into eccDNAs. As suggested by previous studies, MMEJ is implicated in eccDNA biogenesis (Lukaszewicz *et al.*, 2021; Moller *et al.*, 2015; Paulsen *et al.*, 2021; Y. Wang *et al.*, 2021). The precise distribution of microhomologous sequences relative to eccDNA break-points will help better

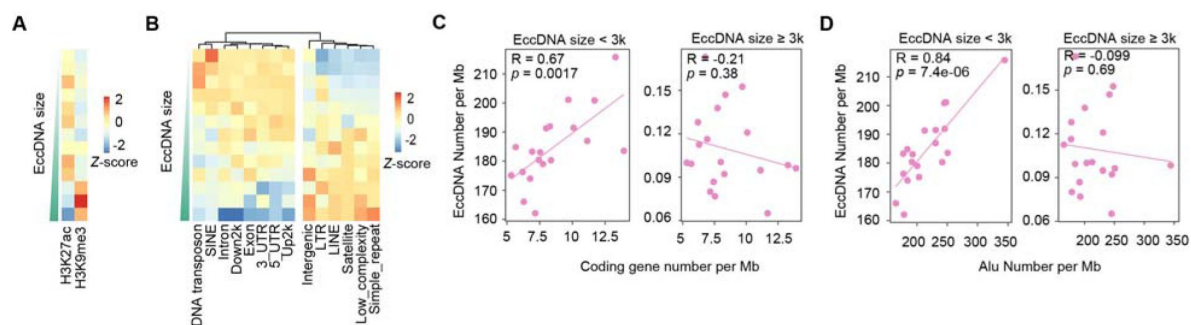


Figure 3.

Large-sized eccDNAs are preferentially generated from heterochromatin regions. **(A)** Distribution at H3K27ac- and H3K9me3-marked regions for eccDNAs of different sizes. **(B)** Distribution at different genomic regions for eccDNAs of different sizes. **(C)** Number of small (<3kb) vs large (≥3kb) eccDNAs per Mb as a function of gene number per Mb. *Pearson* correlation coefficients and two-sided *t*-test *p*-values are indicated. **(D)** Number of small (<3kb) vs large (≥3kb) eccDNAs per Mb as a function of Alu number per Mb. *Pearson* correlation coefficients and two-sided *t*-test *p*-values are indicated.

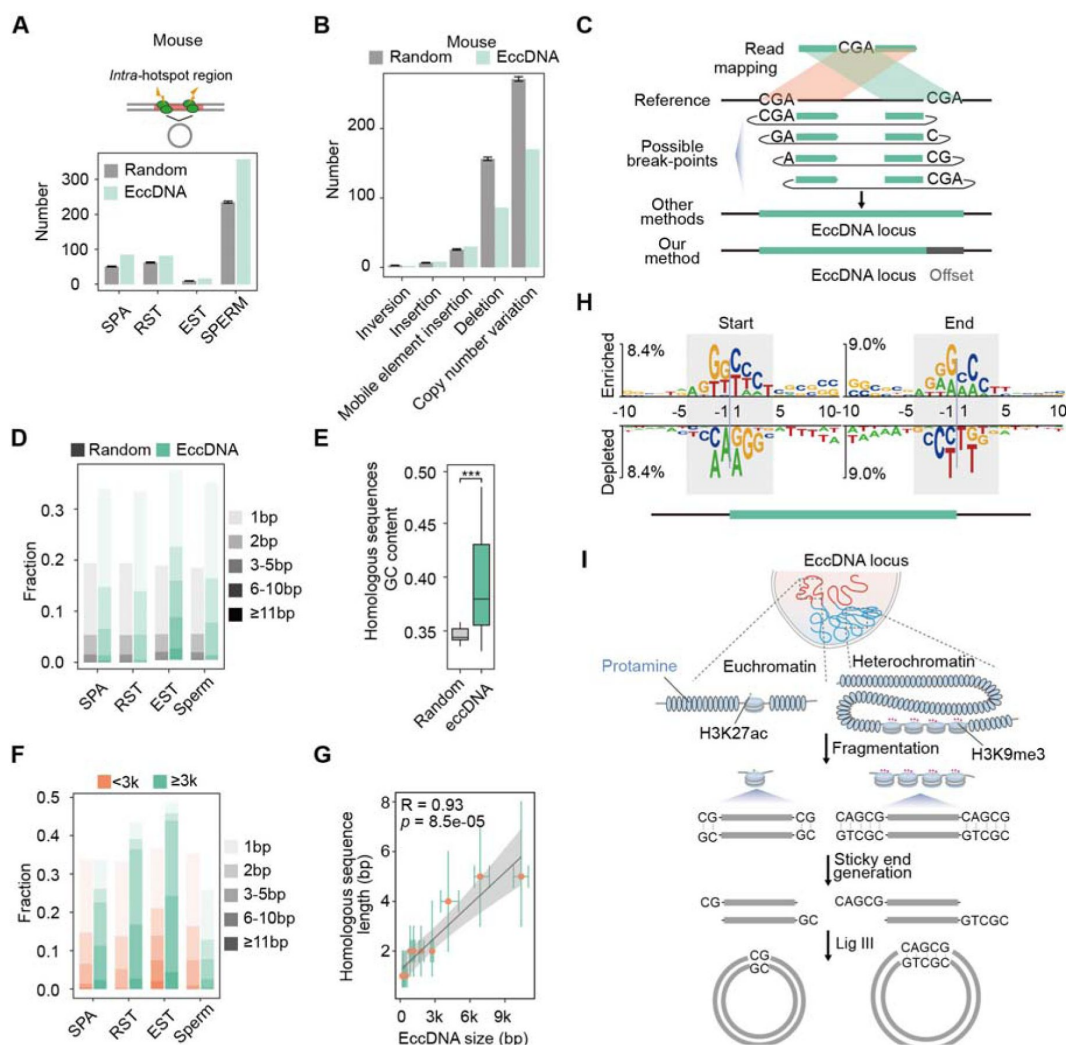


Figure 4.

Microhomology directed ligation accounts for emergence of most eccDNAs. **(A)** Numbers of eccDNAs or randomly-selected control regions overlapped with recombination hotspots in mouse. **(B)** Shown are numbers of mouse sperm eccDNAs or randomly-selected control regions having 95% reciprocal overlap with different types of structural variations. **(C)** Illustrated are how an eccDNA with homologous sequences (CGA) at two ends is identified from short-read sequencing data by our methods vs other methods. **(D)** Percentages of homologous sequences of different lengths (coded by different color saturation levels) are shown for eccDNAs and randomly-selected control regions. **(E)** GC content of homologous sequences and randomly-selected control regions. **(F)** Percentages of homologous sequences of different lengths (coded by different color saturation levels) are shown for small-sized (<3kb) and large-sized (≥3kb) eccDNAs. **(G)** Length of microhomologous sequences as a function of the eccDNA size. Data points are shown as median plus lower (25%) and upper (75%) quartiles. The shaded area is 95% confidence interval of linear regression line. *Pearson* correlation coefficient and two-sided t-test p -value are indicated. **(H)** Sequencing motif analysis for ±10bp leftmost left ends and ±10bp leftmost right ends of eccDNAs with no perfectly matched homologous sequences observed. **(I)** Model for MMEJ-directed eccDNA biogenesis.

understand how and to what extent MMEJ might contribute to eccDNA biogenesis. However, we noted that the presence of microhomologous sequences will hinder precision eccDNA break-point identification (**Figure 4C**), which is not well dealt with by existing methods for eccDNA detection, including ECCsplorer ([Mann et al., 2022](#)), Circle_finder ([Kumar et al., 2017](#)), Circle_Map ([Prada-Luengo et al., 2019](#)) and ecc_finder ([P. Zhang et al., 2021](#)) (**Figure 4—figure supplement 2A**). Short sequencing reads spanning eccDNA break-points will be mapped to the genome as split reads, with its first part mapped to the right end of eccDNA, and the second part to the left end. If the sequence in front of the left eccDNA end is homologous to the right eccDNA end, or if the sequence following the right eccDNA end is homologous to the left eccDNA end, the homologous regions will be included in both parts of split reads to reach to a maximal length of matches, and many existing methods will mistake the eccDNA plus two homologous regions as the whole eccDNA region (**Figure 4C**). Being aware of it, we developed a base-resolution method for eccDNA identification on the basis of previous efforts (**Figure 4—figure supplement 2B**) ([Kumar et al., 2017](#); [Moller et al., 2018](#)). When homologous sequences are present, we record the coordinates of the leftmost form of eccDNA and an offset corresponding to the length of homologous sequences to represent all possible eccDNA variants (**Figure 4C**). Similar to ECCsplorer ([Mann et al., 2022](#)), Circle_finder ([Kumar et al., 2017](#)), Circle_Map ([Prada-Luengo et al., 2019](#)) and ecc_finder ([P. Zhang et al., 2021](#)), our method was not designed to identify eccDNAs that encompass multiple gene loci.

We evaluated the performance of our method in comparison with existing methods. Firstly, we simulated paired-end reads derived from a set of eccDNAs with homologous sequences around breakpoints and employed all methods for eccDNA identification (see Materials and methods). In total, 97.9%, 97.9%, 97.4%, 95.3% and 91.1% eccDNA regions could be detected by our method, Circle_Map, Circle_finder, ecc_finder and ECCsplorer, respectively (**Figure 4—figure supplement 2C**). This result suggest that our method has comparable performance with existing methods in detecting eccDNA regions. Moreover, our method could faithfully assign breakpoints with 97.4% accuracy, in contrast to no more than 15% by other methods (**Figure 4—figure supplement 2D**). Secondly, we applied all methods on one dataset generated in this study. Again, our method had comparable sensitivity and specificity with existing methods (especially Circle_finder and Circle_Map) in detecting eccDNA regions (**Figure 4—figure supplement 2E**). At least 60% of eccDNAs with homologous sequences were misannotated by ECCsplorer, ecc_finder, Circle_finder and Circle_Map, respectively (**Figure 4—figure supplement 2A** and **F**). Overall, our method shows a high efficiency and accuracy in precise eccDNA detection.

In contrast to simulated controls (15%), more than one third of eccDNAs had ≥ 1 bp homologous sequences, most of which were shorter than 5bp (**Figure 4D**), suggesting the involvement of MMEJ in eccDNA biogenesis. The GC content of homologous sequences was higher than that of simulated control regions, permitting stronger base-pairing for efficient MMEJ (**Figure 4E**). Considering that two free-ends of long DNA fragments might be not as spatially close as those of short DNA fragments, formation of longer eccDNA should more rely on longer homologous sequences for stable base pairing. Indeed, large-sized eccDNAs in SPA, RST and EST cells did show higher percentage of ≥ 2 bp homology than small-sized eccDNAs, and large-sized eccDNAs in sperm cells showed higher percentage of > 5 bp homology (**Figure 4F**). A significant positive correlation between lengths of homologous sequences and eccDNA sizes was observed (**Figure 4G**). We further reasoned that for the remaining two thirds of eccDNAs that were lack of perfectly-matched homologous sequences, imperfect homologous sequences might be present. Accordingly, we noted the same sequence motifs between eccDNA starts and sequences following eccDNA ends, and between eccDNA ends and sequences in front of eccDNA starts (**Figure 4H**). Similar observations have been made also by others before ([Sin et al., 2020](#)), however they failed to precisely locate the homologous sequences relative to eccDNA break-points. We propose that sticky ends of DNA fragments with homologous sequences might base pair with each other, and then be ligated by DNA ligase, *e.g.*, DNA ligase III ([Y. Wang et al., 2021](#)), to form eccDNAs (**Figure**

41 [↗](#)). In sum, among all proposed mechanisms ([Chiu et al., 2020](#) [↗](#); [Dillon et al., 2015](#) [↗](#); [Moller et al., 2015](#) [↗](#); [Paulsen et al., 2021](#) [↗](#); [Sin et al., 2020](#) [↗](#)), MMEJ-mediated ligation accounts for emergence of most eccDNAs at least in germline cells.

The eccDNA biogenesis mechanism is conserved in somatic tissues and in human

Upon revealing the major biogenesis mechanism of mouse germline eccDNAs, we further examined whether the mechanism is unique to germline cells or common in somatic tissues. We therefore analyzed public-available eccDNA data from various mouse tissues ([Dillon et al., 2015](#) [↗](#)). Sequence-based prediction revealed significantly higher nucleosome occupancy probability for ~180 bp and ~360bp eccDNA regions, suggesting their origin from oligonucleosomal fragments (**Figure 5—figure supplement 1A** [↗](#)). In contrast to simulated controls (~20%), more than 1/3 of eccDNAs had microhomologous sequences, most of which were shorter than 5bp (**Figure 5—figure supplement 1B** [↗](#)). The remaining 2/3 of eccDNAs had the same sequence motifs between eccDNA starts and sequences following eccDNA ends, and between eccDNA ends and sequences in front of eccDNA starts (**Figure 5—figure supplement 1C** [↗](#)). The genomic distribution of eccDNAs closely matched with that of eccDNAs whose generation was dependent on apoptotic DNA fragmentation (**Figure 5—figure supplement 1D** [↗](#)). Altogether, these results indicate microhomology directed ligation of oligonucleosomal fragments in apoptotic cells significantly contributes to eccDNA biogenesis in different mouse tissues.

We next examined whether the mechanism is also conserved in human by analyzing the features of human sperm eccDNAs detected in this study (see Materials and methods). Similarly, human sperm eccDNAs were originated from oligonucleosomal fragmentation as well, as suggested by the pronounced size periodicity of ~180bp (**Figure 5A** [↗](#)), higher GC content and nucleosome occupancy probability (**Figure 5B** [↗](#) and **C** [↗](#)). We further performed base-level comparison between eccDNAs and meiotic recombination hotspots. Consistent with what we observed in mouse, only about 600 eccDNAs (<0.7% of human sperm eccDNAs) were located in one recombination hotspot or two different hotspots (**Figure 5D** [↗](#)), and only dozens of, or a few hundred eccDNAs in human sperm cells coincided with known genomic deletions (**Figure 5E** [↗](#)). Overall, our analysis disfavors the meiotic recombination to eccDNA biogenesis. Instead, we observed higher frequency of micro-homologous sequences for eccDNAs than simulated control regions (**Figure 5F** [↗](#)). Moreover, large-sized eccDNAs had longer homologous sequences possibly for stable base pairing, as indicated by the strong positive correlation between lengths of homologous sequences and eccDNA sizes (**Figure 5G** [↗](#)). Altogether, these results suggest microhomology directed ligation of nucleosome protected DNA fragments is a conserved pathway for germline eccDNA generation in both human and mouse.

Discussion

The biogenesis mechanisms and biological implications of eccDNAs are big puzzles. Increasingly evidence has emerged to relate eccDNA formation with nucleosome-protected DNA fragments, however, to our knowledge, no study provides direct evidences. The comparison between high number of sperm eccDNAs and retained histones in sperm cells after histone-to-protamine transition ([Torres-Flores et al., 2020](#) [↗](#)) provide a good way to clarify the abovementioned issues. Here, we profiled eccDNA landscape during mouse spermatogenesis, and explicitly linked eccDNAs with nucleosome-protected DNA fragments (**Figure 2** [↗](#)). According to these results, we speculate that any oligonucleosomal fragments might have a chance of eccDNA formation as long as microhomologous sequences are present at the ends. In other words, any single cells might generate distinct sets of eccDNAs and thus the theoretical number of eccDNA is extremely large.

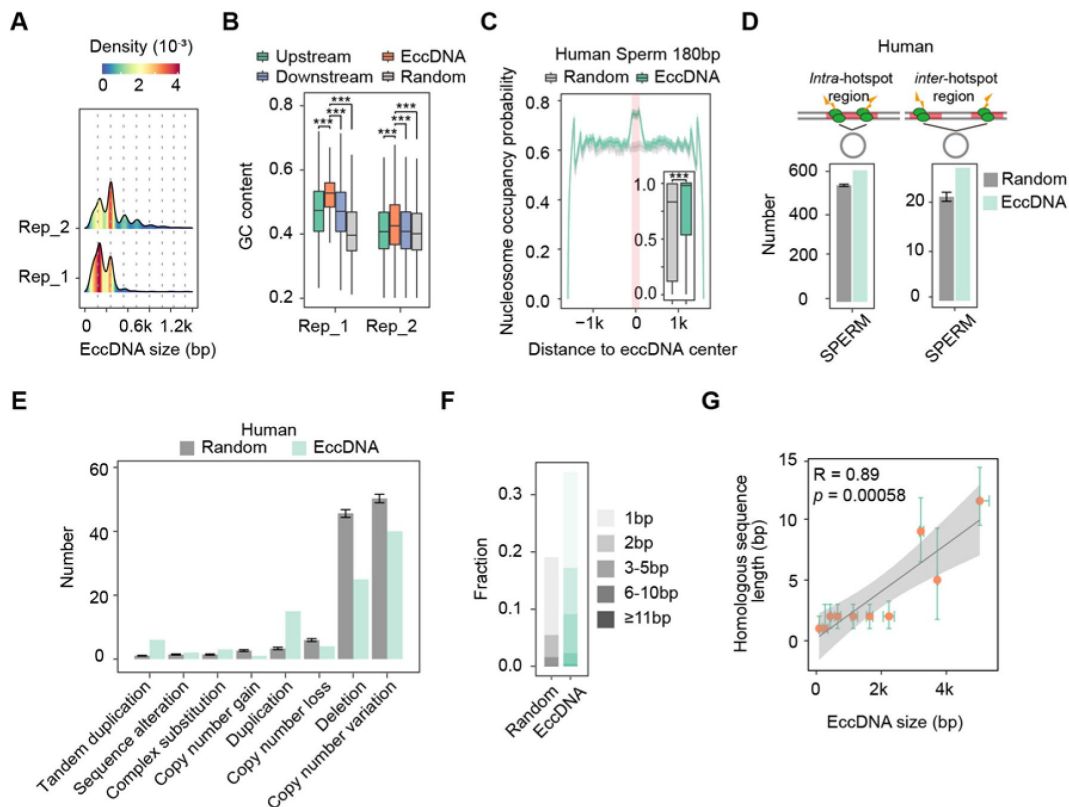


Figure 5.

The biogenesis mechanism of germline eccDNAs is conserved between human and mouse. **(A)** Size distribution of sperm eccDNAs in two biological replicates. **(B)** GC contents of sperm eccDNAs, regions upstream and downstream of eccDNAs, and randomly-selected length-matched control regions. ***two-sided *Wilcoxon* test p -value <0.001. **(C)** Predicted probability of nucleosome occupancy for eccDNA and randomly-selected length-matched control regions (highlighted by red-shaded area), and surrounding regions. Boxplots showing the probability distribution of individual eccDNAs and control regions. ***two-sided *Wilcoxon* test p -value <0.001. **(D)** Numbers of eccDNAs or randomly-selected control regions overlapped with recombination hotspots in human. EccDNAs located completely within a hotspot (Intra-), or with both ends overlapped with two different hotspots (Inter-) are shown separately. **(E)** Shown are numbers of human sperm eccDNAs or randomly-selected control regions having 95% reciprocal overlap with different types of structural variations. **(F)** Percentages of homologous sequences of different lengths (coded by different color saturation levels) are shown for eccDNAs and randomly-selected control regions. **(G)** Length of microhomologous sequences as a function of the eccDNA size. Data points are shown as median plus lower (25%) and upper (75%) quartiles. The shaded area is 95% confidence interval of linear regression line. *Pearson* correlation coefficient and two-sided t -test p -value are indicated.

This might explain why sperm cells, which have a higher starting cell number, have more eccDNAs detected, even though the same amount of eccDNAs (10 ng) for each cell type was used for library preparation and high-coverage sequencing (~150 million reads).

Multiple biogenesis mechanisms of eccDNA have been proposed ([Chiu et al., 2020](#) [Chiu et al., 2020](#); [Dillon et al., 2015](#) [Dillon et al., 2015](#); [Moller et al., 2015](#) [Moller et al., 2015](#); [Paulsen et al., 2021](#) [Paulsen et al., 2021](#); [Sin et al., 2020](#) [Sin et al., 2020](#)), it is however unclear which one plays the dominant role. With our nucleotide-resolution eccDNA break-point detection method, we demonstrated that microhomologous sequences are present at boundaries of most eccDNAs (**Figure 4** [Figure 4](#)), suggesting the MMEJ-mediated eccDNA formation as the major mechanism for eccDNA biogenesis. However, rolling circle amplification in Circle-seq protocol preferentially increases the copy numbers of smaller eccDNAs, and our eccDNA detection method relying on uniquely-mapped reads might favor the detection of small-sized eccDNAs with homologous sequences. It remains to be determined whether these small-sized eccDNAs with microhomologies are the dominant eccDNA species in the native composition. To examine whether the unexplored size populations of eccDNAs by Circle-seq were also associated with microhomologous sequences, we analyzed eccDNA data generated with long-read sequencing ([Henriksen et al., 2022](#) [Henriksen et al., 2022](#)) or amplification-free strategies ([Mouakkad-Montoya et al., 2021](#) [Mouakkad-Montoya et al., 2021](#)). Our sequence feature analyses also revealed the presence of homologous sequences surrounding eccDNA breakpoints (**Figure 5—figure supplement 1E** [Figure 5—figure supplement 1E](#) and **F** [Figure 5—figure supplement 1F](#)), suggesting the involvement of MMEJ-mediated ligation for large-sized eccDNA as well. We further found the biogenesis mechanism of germline eccDNAs is common to eccDNAs in other tissues, and conserved between human and mouse. However, it remains unclear whether eccDNA generation from healthy cells not undergoing apoptosis is also mediated by MMEJ.

Spontaneous germ cell death has been observed during the normal spermatogenesis ([Liu et al., 2017](#) [Liu et al., 2017](#); [Shaha et al., 2010](#) [Shaha et al., 2010](#); [Weinbauer et al., 2001](#) [Weinbauer et al., 2001](#); [Young et al., 2001](#) [Young et al., 2001](#)). Given the periodic size distribution (**Figure 1C** [Figure 1C](#)) and similar genomic features with eccDNAs from apoptotic DNA fragmentation in somatic cells (**Figure 1** [Figure 1](#) and **Figure 4—figure supplement 1** [Figure 4—figure supplement 1](#)), we reason that mouse germline eccDNAs may be also the germ cell death products. As the final place for sperm maturation and storage, epididymis might contain unhealthy sperm cells undergoing cell death and so DNA fragments, which might additionally contribute to the high eccDNA loads of sperm cells than others. It is possible that failure of histone-to-protamine exchange might account for some sperm cell death. It has been shown that H3.3, generally linked to H3K4me3 marks, plays key roles in modulating TP1 removal and PRM1 incorporation for nucleosome eviction and replacement by protamine ([Wang et al., 2019](#) [Wang et al., 2019](#)), and loss of H3.3 will increase the cell death rate ([Yuen et al., 2014](#) [Yuen et al., 2014](#)). H3K36me3 and H3K9me3, whose demethylase is inhibited by H3.3 ([Udugama et al., 2022](#) [Udugama et al., 2022](#)), are also involved in histone replacement ([Wang et al., 2019](#) [Wang et al., 2019](#)), failure of which might underlie the eccDNA generation as well. In line with it, sperm eccDNA regions correspond to depletion of H3.3 variant and H3K4me3, H3K36me3 and H3K9me3 histone modifications (**Figure 2D** [Figure 2D](#) and **E** [Figure 2E](#)). This suggests the number of sperm eccDNAs might be highly associated with cell death, and sperm eccDNA may serve as a clinical biomarker for quality assessment of healthy sperms.

It is still debatable whether meiotic recombination is an important source of eccDNA generation. Both positive and negative correlations between eccDNAs and meiotic recombination rates at chromosomal level have been reported previously ([Henriksen et al., 2022](#) [Henriksen et al., 2022](#); [Lukaszewicz et al., 2021](#) [Lukaszewicz et al., 2021](#)). Our results suggest that this discrepancy can be largely reconciled in the light of two eccDNA groups of different sizes. We noted that small-sized and large-sized eccDNAs are preferentially derived from euchromatin and heterochromatins respectively (**Figure 3** [Figure 3](#)). Since meiotic recombination hotspots are enriched at euchromatin regions ([Lange et al., 2016](#) [Lange et al., 2016](#)), small-sized eccDNAs are therefore positively correlated with recombination rates but the opposite is true for large-sized eccDNAs (**Figure 3—figure supplement 1** [Figure 3—figure supplement 1](#)). Therefore, the observed correlations between eccDNA density and recombination meiotic rate at chromosomal level are simply indirect reflections of their preferences to different chromatin regions.

We further investigated the coincidence between eccDNAs and meiotic recombination hotspots and structural variations at base level, and found that sperm cell eccDNAs does seem to be a major source or by-products of *de novo* deletions, likely because that extensive eccDNA formation might only occur at dead sperm cells that having no chance to be transmitted to the next generation. However, we cannot fully exclude the possibility that some eccDNAs emerging in normal sperm cells and do not affect their viability will be accompanied by *de novo* deletions or re-integrated into genome to create structural variations. Furthermore, they might stimulate innate immune responses (Y. Wang *et al.*, 2021 [↗](#)), server as extracellular vesicles encoding small RNAs (Paulsen *et al.*, 2019 [↗](#)), and be taken as biomarkers (Lv *et al.*, 2022 [↗](#)), which represent exiting research directions.

Methods

Isolation of mouse germ cells

Testes from adult C57BL/6 mice were decapsulated, and the seminiferous tubules were torn into small pieces. After incubating in 8 ml PBS containing 1 mg/ml collagenase (Sigma, C5138, St. Louis, MO, USA) and 1 mg/ml hyaluronidase (Sigma, H3506, St. Louis, MO) at 37°C for 6 min, the dispersed seminiferous tubules and cells were incubated at 37°C for 5 min with gentle shaking. Cells were collected by centrifugation at $200 \times g$ for 5 min at 4°C, and washed once with PBS, resuspended in 15 ml PBS containing 0.25% Trypsin (Gibco, 25200072) and 1 mg/ml DNase I (Sigma, AMPD1), and incubated at 37°C for 5 min with gentle shaking. Thereafter, cells were collected and washed once with PBS. After filtration through a 70 μ m Nylon Cell Strainer (BD Falcon, 352350), the cells were suspended in 30 mL of PBS and stained at 37°C with 5 mg/mL Hoechst 33342 (Thermo Scientific, 62249) for 30 min. Hoechst fluorescence were detected with a 450-nm band-pass filter for blue fluorescent (Hoechst Blue) or a 675-nm band-pass filter for red fluorescent (Hoechst Red). Mouse spermatocytes, round spermatids and elongating spermatids were sorted by BD FACSAria™ Fusion (BD Biosciences). Mouse mature spermatozoa were collected form mouse cauda epididymis. The cauda epididymis was quickly cut into pieces and incubated in 1 ml pre-warmed human tubal fluid (HTF) (Millipore, MR-070-D) for 15 min at 37°C, thus allowing the mature spermatozoa to release from the tissue. After filtration through a 40 μ m Nylon Cell Strainer (BD Falcon, 352340), mature spermatozoa were collected and washed three times with PBS. About 10^6 SPA, RST and EST cells and 10^7 sperm cells were obtained and used for the eccDNA extraction.

Immunofluorescence

Mouse germ cells were spread on glass slides that were air dried. The slides were fixed in 4% PFA at room temperature for 5 min, washed with PBS three times. After blocking with 5% bovine serum albumin, the primary antibodies (rabbit anti-SYCP3 pAb (Abcam, ab15093, 1:400); mouse anti- γ H2AX pAb (Millipore, 05-636, 1:400); Mouse anti-sp56 mAb (QED Bioscience, 55101, 1:200); Rabbit Tubulin pAb (ABclonal, AC007, 1:100)) were added to the slides, followed by 14-16 h incubation at 4°C. After washing the slides with PBS, the secondary antibodies were added, followed by 1.5 h incubation. The cell nuclei were stained with DAPI for 5 min. Images were observed using a fluorescence microscope SP8 microscope (Leica).

Human adult sperm sample preparation

The sperm donation candidates in the present study were healthy young Chinese men. Each candidate completed a medical examination and extensive medical/social questionnaire to exclude any potential individuals with genetic or major medical problems (such as cardiovascular diseases and sexually transmitted diseases) listed in the Basic Standard and Technical Norms of Human Sperm Bank published by Chinese Ministry of Health. Smokers, drug abusers, and heavy drinkers were also excluded. The rest of the candidates signed a voluntary sperm donation

informed consent and agreed to live in Beijing for at least 6 months. The sperm bank also recorded the candidates' age, date of birth, and date of semen collection. The ethical approval in this study were provided by the Reproductive Study Ethics Committee of Peking University Third Hospital (2017SZ-035). Semen samples were selected through 40% density gradient of PureSperm (Nidacon International, Molndal, Sweden) by centrifugation (500g, 30 min) at room temperature and washed with phosphate-buffered saline (PBS) for three times; the obtained spermatozoa were used for the eccDNA extraction.

Circle-Seq

Purification of mouse and human eccDNAs was performed as previously described, with minor modifications (Moller *et al.*, 2018). In brief, samples were resuspended in 500LJ μ l Lysis solution (10mM Tris pH7.4, 100mM NaCl, 1% SDS, 1% Sarkosyl, 150mM DTT) with 10 μ L Proteinase K (Thermo Scientific, EO0491), and incubated overnight at 55LJ°C. After cell lysis, phenol: chloroform: isoamyl alcohol was added and mixed, centrifuged at 13,000 g for 15min at 4LJ°C. The supernatant was moved to a new tube, incubated with 500 μ L isopropanol at room temperature for 10 min, and centrifuged at 13,000 g for 15min at 4LJ°C. The resulted DNA pellet was washed with 1 ml 70% ethanol, treated with alkaline to separate chromosomal DNA from eccDNAs by rapid DNA denaturing–renaturing, followed by column chromatography on an ion exchange membrane column (TIANprep Mini Plasmid Kit, DP103). Column-bound DNA was eluted in TE buffer (10 mM Tris-Cl, pH 8.0; 1 mM EDTA, pH 8.0) and treated with AsiSI (NEB, R0630S) and PacI (NEB, R0547S) endonucleases at 37 °C for genomic DNA and mtDNA fragmentation. The remaining linear DNA was treated by exonuclease (Plasmid-Safe ATP-dependent DNase, Lucigen, E3101K) at 37LJ°C for 1 week, during which additional ATP and DNase was added every 24LJh (30 units per day) according to the manufacturer's protocol (Plasmid-Safe ATP-dependent DNase, Lucigen, E3101K). The eccDNA samples were cleaned by phenol: chloroform: isoamyl alcohol once, followed by ethanol precipitation. EeccDNAs were then amplified by phi29 polymerase (NEB, M0269L) at 30LJ°C for 16LJh. Phi29-amplified DNA was cleaned by phenol: chloroform: isoamyl alcohol once, followed by ethanol precipitation. The DNA samples were sonicated to a set size of 250 bp with an M220 Focused-ultra sonicator (Covaris, Woburn). Sequencing libraries were generated with NEBNext® UltraTM II DNA Library Prep Kit for Illumina (NEB, E7645S), according to the manufacturer's instructions. 10 ng of eccDNA samples were used for library preparation. NEBNext Multiplex Oligos for Illumina (Set 1, NEB #E7600) were used for PCR amplification of adaptorLligated DNA. Libraries were purified with SPRIselect® Reagent Kit (Beckman Coulter, Inc #B23317). PairedLJend 150 bp sequencing was performed on Illumina NovaSeq 6000 System.

Quality Control of Circle-seq and eccDNA validation

Exogenous circular DNA (pUC19) and mtDNA were measured by qPCR in a QuantStudio 6 Flex Real-Time PCR System, following the protocol of the manufacturer. The primer probes used for qPCR included: pUC19 forward: 5'-AGC GAA CGA CCT ACA CCG AAC-3', pUC19 reverse: 5'-CTC AAG TCA GAG GTG GCG AAA C-3'; MTND2 forward: 5'-AAC AAA CGG TCT TAT CCT TAA CAT AAC A-3', MTND2 reverse: 5'-TGG GAT CCC TTG AGT TAC TTC TG-3'; H19 forward: 5'-GTA CCC ACC TGT CGT CC-3', H19 reverse: 5'-GTC CAC GAG ACC AAT GAC TG-3'. EccDNA validation was performed by outward PCR in genomic DNA and eccDNA extractions. The primer probes used for eccDNA validation included: Clone 1 (chr1:73132172-73132640) in-forward: 5'-TTT TCC TGG AGC ACA CTA GC-3'; Clone 1 in-reverse: 5'-CAT GCT AAA CAA AGC ATG TCA C-3'; Clone 1 out-reverse: 5'-CAA CTG ACA CCA ACC ACA TC-3'; Clone 2 (chr1:120058476-120058826) in-forward: 5'-CCT GCC ACT GCT CTG CAT TC-3'; Clone 2 in-reverse: 5'-AGA TGC AAT AGG ACC AGG ATG-3'; Clone 2 out-reverse: 5'-GCC CAG AGC AGA ATC CAA AG-3'; Clone 3 (chr13:95405916-95406337) in-forward: 5'-GGT CAC ACA TGC AAA TGT CC-3'; Clone 3 in-reverse: 5'-AAC ATA CCT GAG ACC CTA GG-3'; Clone 3 out-reverse: 5'-TTC CCA CAG CTA TGC TCA GC-3'.

EccDNA detection

We developed a nucleotide-resolution eccDNA detection pipeline on the basis of previous efforts (**Figure 4—figure supplement 1B**) (Kumar *et al.*, 2017; Moller *et al.*, 2018). Briefly, SeqPrep (<https://github.com/jstjohn/SeqPrep>) was used to trim adapter sequences and merge the overlapping paired-end reads into singleton long reads, followed by reads mapping to GRCm38 reference genome using BWA MEM version 0.7.17-r1188 (Li *et al.*, 2009). Samblaster version 0.1.26 (Faust *et al.*, 2014) or an in-house Perl script was used to remove PCR duplicates and separate alignments into split reads, discordant and concordant reads. Candidate eccDNAs were firstly identified based on split reads (high-confidence ones). If the total length of two sub-alignments of split reads exceeded the read length, homologous sequences were searched. When homologous sequences were found, we recorded the coordinates of the leftmost form of eccDNA and an offset corresponding to the length of homologous sequences to represent all possible eccDNA variants. Potential split reads that failed to be mapped as split reads in the first place (low-confidence ones) as well as discordant reads were identified and counted using in-house Perl scripts. The average coverages (in terms of RPK) for candidate eccDNAs and surrounding regions were then calculated based on all different types of reads. Any eccDNA supported by at least two high-confidence split reads or discordant reads, with its 95% region covered by at least one read, and with its average coverage twice of that of its surrounding region, is considered as a high-confidence eccDNA.

Evaluation of our EccDNA detection method

We randomly selected 1,000 eccDNAs containing ≥ 2 bp of microhomology sequences from this study. Ten copies of each eccDNA sequence were concatenated to mimic the product of rolling amplification process. We then randomly extracted 50 fragments ranging from 250 to 350 bp from each concatenated sequence to mimic the sonication process, and generated 150 bp paired-end reads from each fragment. We applied Circle_finder, Circle_Map, ecc_finder, ECCsplorer and our own method to identify eccDNAs from the simulated paired-end reads. Detected eccDNAs with at least 95% reciprocal similarity with one of the 1,000 eccDNAs were considered positive hits. We considered eccDNA boundaries to be correctly assigned only if they had the same start and end coordinates.

EccDNA characterization

Gene structure annotations in mouse and human are based on Ensembl GRCm38 Release 102 and Ensembl GRCh38 Release 104, respectively. Repeat elements are annotated by RepeatMasker database. Random region generation, sequence composition calculation, and overlapping region determination were all done with Bedtools version 2.30.0 (Quinlan *et al.*, 2010). The nucleosome occupancy probability of the eccDNAs and the surrounding regions was predicted using predNuPoP function of NuPoP version 2.2.0 R package (Xi *et al.*, 2010). Motif analysis and visualization were done by Two Sample Logo web server (Vacic *et al.*, 2006). Public ChIP-seq datasets (Jung *et al.*, 2019; Jung *et al.*, 2017; Singh *et al.*, 2021) were aligned to GRCm38 reference genome using Bowtie, and sorted and indexed using SAMtools version 1.7 (Danecek *et al.*, 2021). Picard MarkDuplicates version 2.18.14 was used to remove PCR duplicates. Peak calling was done with Macs2 version 2.2.7.1 (Y. Zhang *et al.*, 2008). Coverage file was generated from BAM file and visualized using deeptools version 3.5.1 (Ramirez *et al.*, 2016).

Data and software access

The raw sequencing data reported in this study have been deposited in the Genome Sequence Archive (Chen *et al.*, 2021) in National Genomics Data Center (Members *et al.*, 2022) (GSA accession numbers: CRA008015 and HRA002957) that are publicly accessible at

Competing interest statement

The authors declare no competing financial interests.

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Author contributions

J.-Y.C., W.L., C.L. and H.J. conceived and designed the experiments. C.L. and Z.Z. performed the majority of the experiments with the help from S.X., Y.C., Y.C. and J.W., J.H. was responsible for the processing and analysis of most genomic data. J.-Y.C. interpreted the data and wrote the paper.

Figure supplements

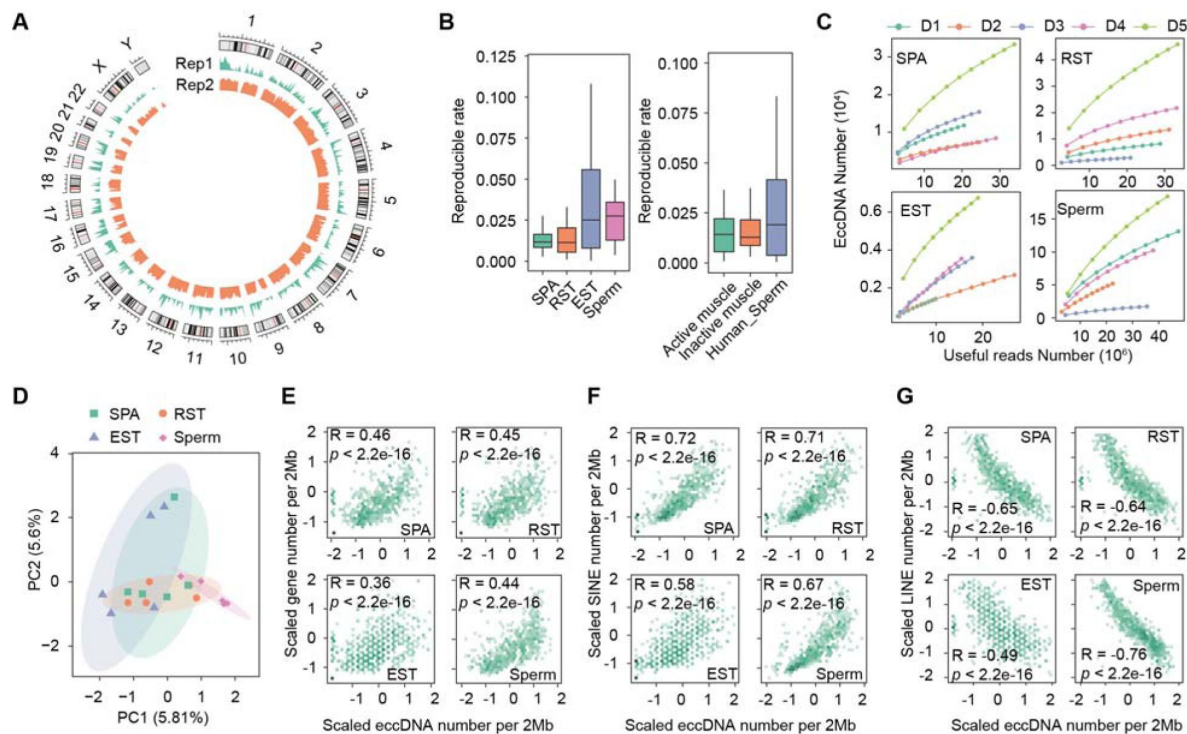


Figure 1—figure supplement 1.

Detection and characterization of eccDNAs. **(A)** Genome-wide distribution of eccDNAs detected in two human sperm samples. **(B)** Histograms showing distributions of reproducible rates between any two replicates of the same biological sample. Left: data generated in this study; Right: public datasets [Moller et al, Nat Commun (2018) 9:1069; Henriksen et al, Mol Cell (2022) 82: 209-217.e7]. **(C)** Number of detected eccDNAs as a function of non-redundant uniquely-mapped reads (useful reads) randomly sampled from indicated Circle-seq sequencing libraries. **(D)** Principal component analysis of biological replicates using Jaccard indexes with all other replicates as features. 75% confidence ellipses are indicated. **(E-G)** eccDNA number per 2 mega bases as a function of gene density **(E)**, density of SINE **(F)** and LINE **(G)** elements.

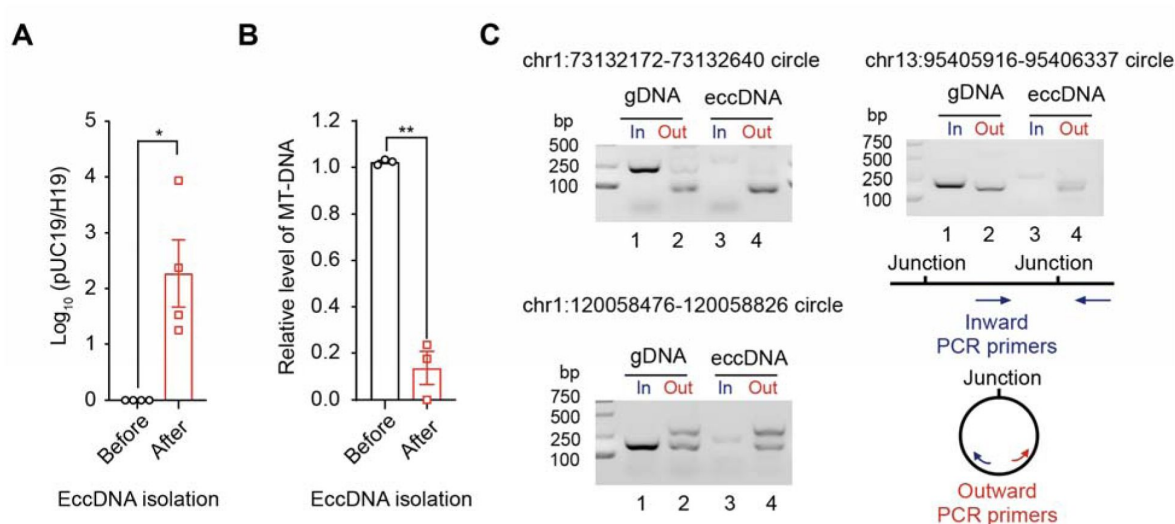


Figure 1—figure supplement 2.

Quality control of the eccDNA isolation procedure. **(A)** qPCR quantification of the ratio of an exogenous circular DNA (pUC19) to a linear DNA locus (H19 gene) before and after eccDNA isolation procedures. The ratio before eccDNA isolation is normalized to 1 (n=4 independent experiments, two-tailed Student's t-test p -value=0.0327). Data are presented as means $\pm$ SEM. *two-tailed Student's t-test p -value < 0.05. **(B)** Relative abundance of mtDNA before and after eccDNA isolation procedures. The abundance of mtDNA before eccDNA isolation is normalized to 1 (n=3 independent experiments, two-tailed Student's t-test p -value=0.0068). Data are presented as means $\pm$ SEM. **two-tailed Student's t-test, p -value < 0.01. **(C)** Gel image showing PCR validation of three eccDNAs using inward and outward PCR primers.

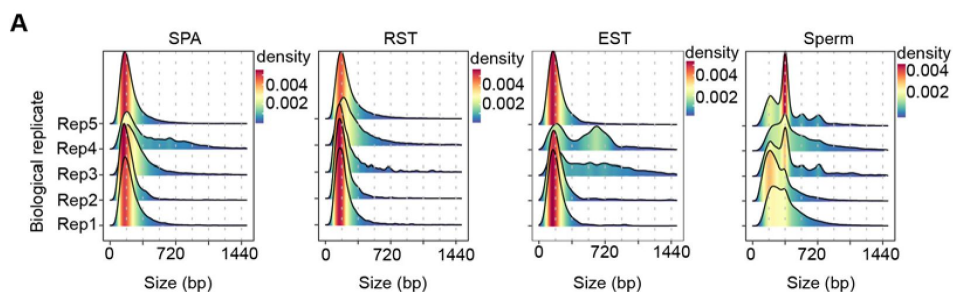


Figure 1—figure supplement 3.

Length distribution of eccDNAs in different cells. Dotted lines indicate multiples of 180 bp.

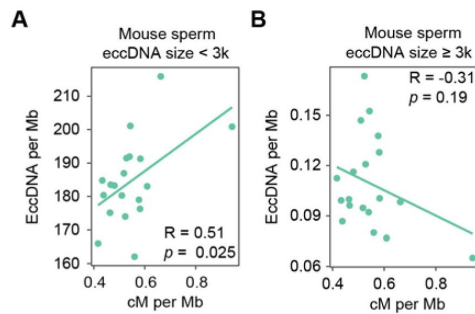


Figure 3—figure supplement 1.

Correlation between the density of small-sized (**A**) vs large-sized (**B**) eccDNAs and the meiotic recombination rate [Jensen-Seaman et al., *Genome Res* (2004) 14, 528-538].

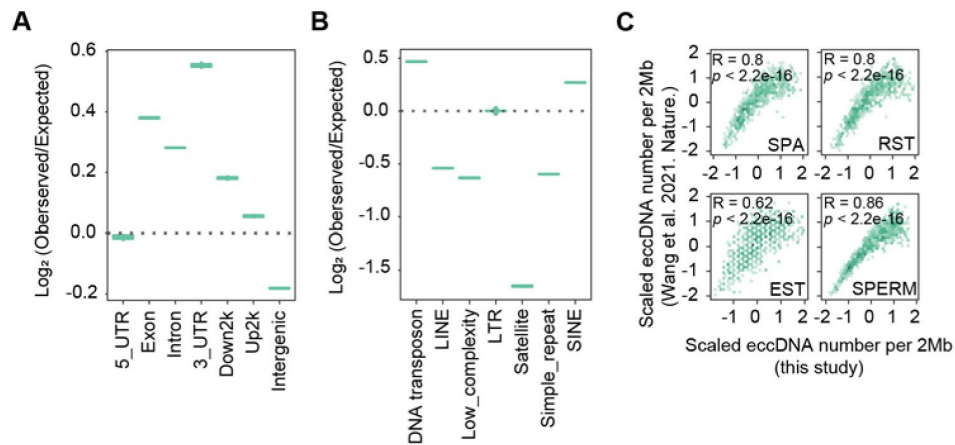


Figure 4—figure supplement 1.

High similarity between sperm eccDNAs detected in this study and those from apoptotic DNA fragmentation reported previously. (**A**) Enrichment of eccDNAs reported in Nature (2021) 599: 308-314 at given genomic regions relative to randomly-selected control regions. (**B**) Enrichment of eccDNAs reported in Nature (2021) 599: 308-314 at given repeat elements relative to randomly-selected control regions. (**C**) EccDNA numbers per 2Mb region in this study as a function of eccDNA numbers per 2 Mb reported in Nature (2021) 599: 308-314.

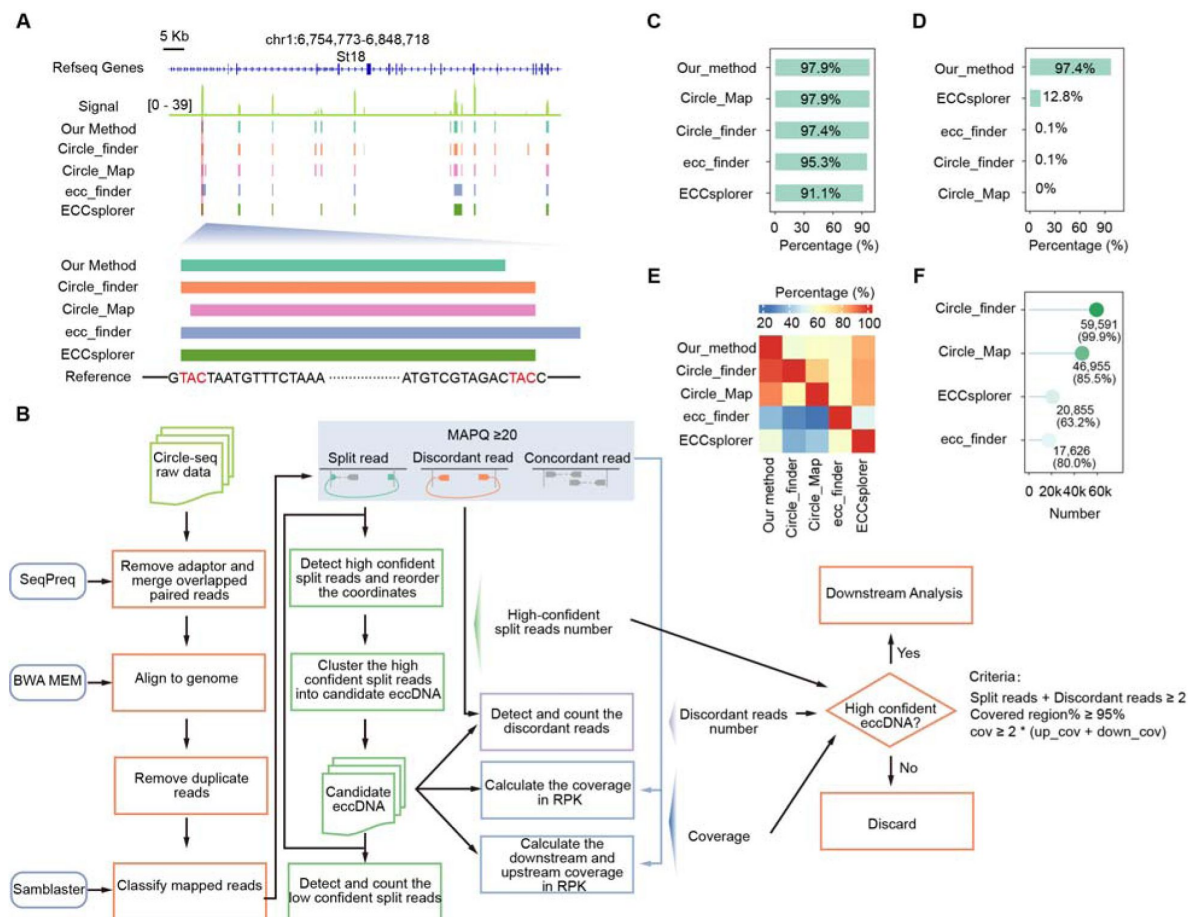


Figure 4—figure supplement 2.

Evaluation of our nucleotide-resolution eccDNA detection method. **(A)** A representative genomic locus showing Circle-seq signals and eccDNAs detected by individual methods. Enlarged are the first eccDNA as well as its sequence information. Homologous sequences (TAC) at the eccDNA end are highlighted in red. **(B)** Schematic representation of our eccDNA detection pipeline. **(C)** Percentages of simulated eccDNA regions correctly detected by different methods. **(D)** Percentages of eccDNA breakpoints correctly assigned by different methods. **(E)** Percentages of eccDNAs detected by individual methods (x-axis) that were co-detected by other methods (y-axis). **(F)** Numbers and percentages of eccDNA breakpoints misassigned by different methods.

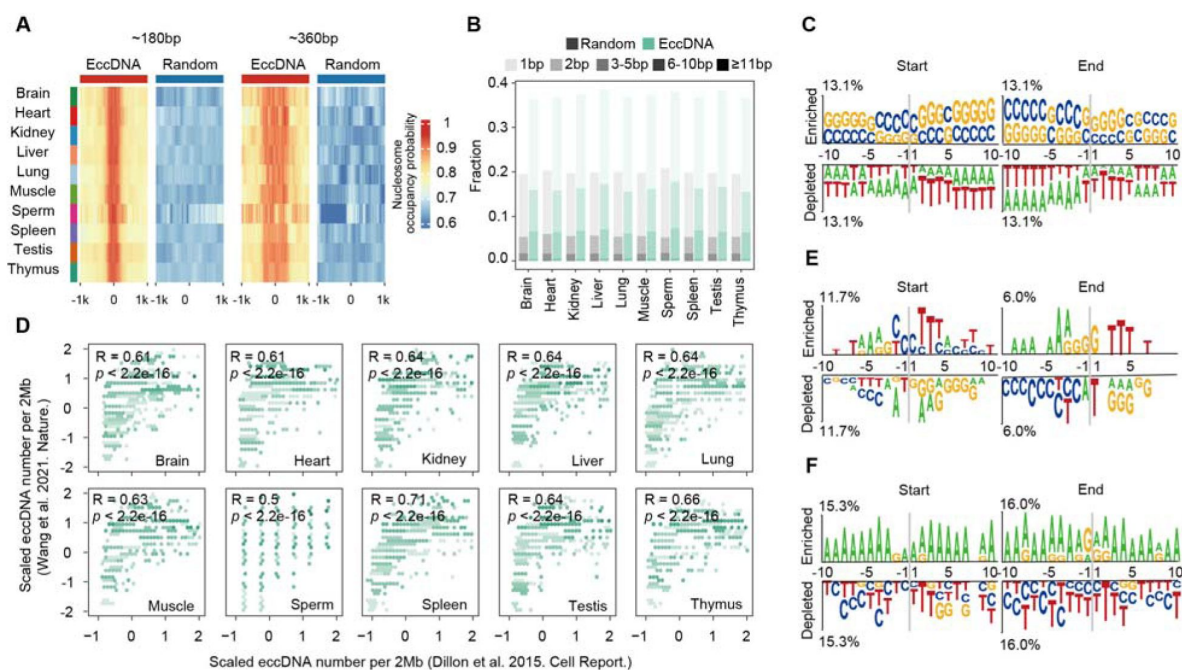


Figure 5—figure supplement 1.

Biogenesis mechanism of eccDNAs in mouse somatic tissues and human sperms. **(A)** Predicted probabilities of nucleosome occupancy for ~180bp and ~360 bp eccDNAs in various mouse tissues. **(B)** Percentages of homologous sequences of different lengths (coded by different color saturation levels) are shown for eccDNAs and randomly-selected control regions. **(C)** Sequencing motif analysis for ±10bp leftmost left ends and ±10bp leftmost right ends of eccDNAs with no perfectly matched homologous sequences observed. **(D)** EccDNA numbers per 2Mb region in this study as a function of eccDNA numbers per 2 Mb reported in Nature (2021) 599: 308-314. **(E and F)** Sequencing motif analysis for ±10bp leftmost left ends and ±10bp leftmost right ends of eccDNAs reported in Mol Cell (2022) 82(1): 209-217 **(E)** and PNAS (2021) 118(47) **(F)**.

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Joint Public Review:

This study investigated the mechanisms and biological processes associated with eccDNA generation in germline cells. They enriched eccDNA from cells at each step of spermatogenesis in mouse as well as human sperm using a commonly-used method to enrich

small eccDNA: column purification, exonuclease digestion, rolling circle amplification, followed by short-read Illumina sequencing. From the fragment size analyses, dominant sizes were shown to be those protected by mono- or di-nucleosomes. The authors developed a computational pipeline to investigate eccDNA breakpoints in detail from split reads and reported a prevalence of a microhomology-mediated mechanism. Features of small germline eccDNA closely matched with small eccDNA generated by apoptosis, suggesting apoptotic germline cells as a major source.

Combined with analyses of publically available data from mouse tissues, the study established a strong link between small eccDNA and DNA fragments protected by mono- or di-nucleosomes. The rigorous investigation of microhomologies revealed that eccDNA sizes correlated with the lengths of microhomology in spermatogonial cells. Small eccDNA tends to originate from euchromatic regions, while longer eccDNA is derived from heterochromatic regions. These are novel findings.

The authors repeatedly stated the rare association between eccDNA and recombination hotspots. The argument was backed by (1) the abundance of eccDNA coming from dead cells, and (2) the small number of eccDNA from SPA cells undergoing meiosis. The argument seems to have a point; however, the observation that the authors recovered hundreds of eccDNA at recombination hotspots may indicate that mitotic recombination is a significant source of eccDNA. Because of the bulk isolation of eccDNA, those eccDNAs were outnumbered by the abundant eccDNA coming from apoptotic cell death. Indeed, eccDNAs from recombination hotspots are slightly more than random in all cell types (Fig. 4A).

Related to this issue, the dominance of both 180- and 360-bp fragments in most mouse tissues put the single 180-bp peaks of SPA, RST, and EST eccDNA in a peculiar position. Fewer numbers of these cells were used for eccDNA isolation than sperm cells, resulting in fewer eccDNA in these cell types, despite the same amount (10ng) of DNA input for rolling circle amplification. There might be a technical issue behind the peculiar observation, which is understandable given the challenging nature of isolating pure cell populations.

Author Response

The following is the authors' response to the original reviews.

Thank you for submitting your article "Microhomology-Mediated Circular DNA Formation from Oligonucleosomal Fragments During Spermatogenesis" for consideration by eLife. Your article has been reviewed by 2 peer reviewers, and the assessment has been overseen by a Reviewing Editor and Diane Harper as the Senior Editor.

eLife assessment

This study provides valuable information on the biogenesis of eccDNAs during spermatogenesis, i.e., eccDNAs in spermatogenic cells are not derived from mitotic recombination hotspots but represent oligonucleosomal DNA fragments from apoptotic male germ cells, whose ends are ligated through microhomology-mediated end-joining. The study is currently incomplete because the method of bioinformatics needs more details and data interpretation should take the amplification bias into consideration.

We highly appreciate the positive assessment of our manuscript. Following the insightful suggestions by editors and two reviewers, we have fully addressed two major concerns, i.e., the missing of method detail and the biased data interpretation.

First, to provide the detail of our bioinformatics methods, i) We have illustrated the principle and steps of our eccDNA detection method by Figure 4C and Figure 4-figure supplement 2B, and submitted our source codes to GitHub (website); ii) We compared the performance of our methods in comparison with four established bioinformatics tools on both simulated and real datasets, and revealed that it has comparable sensitivity and specificity (Figure 4—figure supplement 2C and E), and much higher accuracy on the assignment of eccDNA boundaries (Figure 4—figure supplement 2A, D and F); and iii) we have added more description to help readers to better understand our method (see Methods – eccDNA Detection).

Second, the amplification bias is indeed a problem of Circle-seq. Following editors' and Reviewer #1's insightful suggestions, we analyzed other datasets generated by amplification-free strategies (Mouakkad-Montoya et al., PNAS, 2021) and long-read sequencing (Henriksen et al., Mol Cell, 2022). We identified the presence of homologous sequences surrounding eccDNA breakpoints in both datasets (Figure 5-figure supplement 1E and F), suggesting the involvement of MMEJ-mediated ligation for the unexplored size populations of eccDNAs by Circle-seq as well. We have discussed this point and added one section to remind readers of the limitations of rolling-circle amplification-based Circle-seq (the 2nd paragraph of Discussion section).

For your and reviewers' convenience, all changes in the revised manuscript have been marked in red. We hope the modified manuscript addresses your and the reviewers' concerns satisfactorily and is suitable for publication in eLife now.

Reviewer #1 (Public Review):

This study aims to address the mechanism of eccDNA generation during spermatogenesis in mice. Previous efforts for cataloging eccDNA in mammalian germ cells have provided inconclusive results, particularly in the correlation between meiotic recombination and the generation of eccDNA. The authors employed an established approach (Circle-seq) to enrich and amplify eccDNA for sequencing analyses and reported that sperm eccDNA is not associated with meiotic recombination hotspots. Rather, the authors reported that eccDNAs are widespread, and oligonucleosomal DNA fragments from sperm undergoing apoptosis, with the ligation of DNA ends by microhomology-mediated end-joining, would be a major source of eccDNA.

The strength of the study includes evaluating the eccDNA contents not only in sperm but also from earlier stages of cells in spermatogenesis. The differences in eccDNA size peaks between sperm and other progenitors, in particular, the unique peak in sperm around 360 bp, are intriguing. Results from sequencing data analysis were presented elegantly.

We are grateful to Reviewer #1 for his or her recognition of the strength of this study.

I also have critiques. First, the lack of eccDNA quality control step is a concern. Previous studies employed electron microscopy to ensure that DNA species are mostly circular before rolling-circle amplification. Phi29 polymerase is widely used for DNA amplification, including whole genome amplification of linear chromosomal DNA. Phi29 polymerase has a high processivity and strand displacement activity. When those activities occur within a molecule, it creates circular DNA from linear DNA in vitro. In vitro-created eccDNA from linear DNA would be randomly distributed in the genome, which may explain the low incidence of common eccDNA between replicates. Therefore, it will be crucial to show that DNA prior to amplification is dominantly circular. Electron microscopy would be challenging for the study because the relatively small number of cells were processed to enrich eccDNA. An alternative method for quality controls includes spiking samples with linear and circular exogenous DNA and measuring the

ratios of circular/linear control DNA before and after column purification/exonuclease digestion. eccDNA isolation procedures can be validated by a very high circular/linear control DNA ratio.

We greatly appreciate Reviewer #1's valuable suggestions. We have introduced an exogenous circular DNA (pUC19) into our samples and measured its abundance relative to a linear DNA locus (H19 gene) before and after eccDNA isolation procedures according to Reviewer #1's suggestion. As anticipated, we observed significant enrichment of pUC19 following eccDNA isolation (new Figure 1-figure supplement 2A). These results affirm the high selectivity of our protocol in enriching eccDNAs.

Another critique is regarding the limitation of the study. It is important to remind the readers of the limitations of the study. As the authors mentioned, rolling circle amplification preferentially increases the copy numbers of smaller eccDNA. Therefore, the native composition of eccDNA is skewed. In addition, the candidate eccDNAs are identified by split reads or discordant read pairs. The details of the mapping process are unclear from the methods, but such a method would require reads with high mapping quality; the identification of eccDNA is expected to require sequencing reads that are mapped to genomic locations uniquely with high confidence, and reads mapped to more than one genomic location, such as highly similar repeat sequences or duplications, are eliminated. Such identification criteria would favor eccDNA formed by little or no homology at the junction sequences, and eliminate eccDNA formed by long homologies at the ends, such as eccDNA formed exclusively by satellite DNA. Therefore, it is not surprising that the authors found the dominance of microhomology-mediated eccDNA. It remains to be determined whether small eccDNA with microhomologies are the dominant species of eccDNA in the native composition. In this regard, it is noted that similar procedures of eccDNA enrichment (column purification, exonuclease digestion, and rolling circle amplification) revealed variable sizes and characteristics of eccDNA in sperm (human from Henriksen et al. or mice from this study), dependent on the methods of sequencing (long-read or short-read sequencing). Considering these limitations, the last sentence of the introduction, "We conclude that germline eccDNAs are formed largely by microhomology mediated ligation of nucleosome protected fragments, and barely contribute to de novo genomic deletions at meiotic recombination hotspots" needs to be revised.

We thank Reviewer #1 for bringing attention to the limitations of the study. Since rolling circle amplification preferentially increases the copy numbers of smaller eccDNA, the exact size distribution of eccDNA in native composition is yet to be determined. As pointed out by Reviewer #1, our mapping and eccDNA detection processes might indeed introduce some biases since we only focused on uniquely-mapped reads. We have addressed and incorporated Reviewer #1's perspectives in our revised manuscript, as detailed in the 2nd paragraph of Discussion section.

Despite these limitations, microhomology mediated ligation of DNA fragments seems to be the major mechanism of eccDNA biogenesis nonetheless. We analyzed eccDNA datasets generated through long-read sequencing (Henriksen et al., Mol Cell, 2022) or amplification-free strategies (Mouakkad-Montoya et al., PNAS, 2021). Although these eccDNAs represented size populations that were largely missed by this study, our sequence feature analyses also revealed the presence of homologous sequences surrounding eccDNA breakpoints, as depicted in the newly added Figure 5-figure supplement 1E and F. Considering that we could not totally overcome these biases in this study, we have toned down some statements and revised the last sentence of the introduction as follows: "We conclude that germline eccDNAs are likely formed by microhomology mediated ligation of nucleosome-protected fragments, and barely contribute to de novo genomic deletions at meiotic recombination hotspots."

Small eccDNA (microDNA) data from various mouse tissues are available from the study by Dillion et al., (Cell Reports 2015). Authors are encouraged to examine whether the notable findings in this study (oligonucleosomal-sized eccDNA peaks and the association with apoptotic cell death) are unique to sperm or common in the eccDNA from other tissues.

We are thankful to Reviewer #1 for this suggestion. We analyzed eccDNA data from various mouse tissues (Dillion et al., Cell Rep, 2015) to see whether our findings are unique to sperm or common for other tissues. Sequence-based prediction revealed significantly higher nucleosome occupancy probability for ~180 bp and ~360bp eccDNA regions, suggesting their origin from oligonucleosomal fragments (Figure 5-figure supplement 1A). In contrast to simulated controls (~20%), more than 1/3 of eccDNAs had microhomologous sequences, most of which were shorter than 5bp (Figure 5-figure supplement 1B). The remaining 2/3 of eccDNAs had the same sequence motifs between eccDNA starts and sequences following eccDNA ends, and between eccDNA ends and sequences in front of eccDNA starts (Figure 5-figure supplement 1C). The genomic distribution of eccDNAs closely matched with that of eccDNAs whose generation was dependent on apoptotic DNA fragmentation (new Figure 5-figure supplement 1D). Altogether, these results indicate microhomology directed ligation of oligonucleosomal fragments in apoptotic cells significantly contributes to eccDNA biogenesis in different mouse tissues. We have described this part in the revised manuscript (see the last 2nd paragraph of Results section).

Reviewer #2 (Public Review):

This study presents a useful investigation of eccDNAs in spermatogenesis of mouse. It provides evidence about the biogenesis of eccDNAs and suggests that eccDNAs are derived from oligonucleosomal DNA fragmentation during apoptosis by MMEJ and may not be the direct products of germline deletions. However, the method of data analyses were not fully described and data analysis is incomplete. It provides additional observations about the eccDNA biogenesis and can be used as a starting point for functional studies of eccDNA in sperms. However, many aspects about data analyses and data interpretations need to be improved.

We thank Reviewer #2 for his or her critical reading. We have provided more method details, performed additional analyses and made some clarifications in our revised manuscript (see below).

- *Most of the conclusions made by the work are only based on the bioinformatics analyses, the validation of these findings using other method (biochemistry/molecular biology method) are missing. For example, no QC results presented for the eccDNA purification, which may show whether contaminants such as linear DNA or mitochondria DNA have been fully removed. Additionally, it is also helpful to use simple PCR to test the existence of identified eccDNAs in sperm or other samples to validate the specificity of the Circle-seq method.*

Following both this Reviewer's and Reviewer #1's suggestions, we performed quality control of eccDNA purification. First, we introduced an exogenous circular DNA (pUC19) into our samples and measured its abundance relative to a linear DNA locus (H19 gene) before and after eccDNA isolation procedures. As anticipated, we observed significant enrichment of pUC19 following eccDNA isolation (Figure 1-figure supplement 2A). Second, mitochondria DNA is supposed to be cleaved into linear DNA by PacI and degraded by exonuclease. As expected, the abundance of mitochondria DNA significantly decreased after eccDNA isolation

procedures (Figure 1-figure supplement 2B). Third, we performed PCR using outward primers and validated three randomly-selected eccDNAs (Figure 1-figure supplement 2C).

• *The reliability of the data analysis methods is uncertain, as the authors constructed and utilized their own pipeline to identify eccDNAs, despite the availability of established bioinformatics tools such as ECCsplorer, eccFinder, and Amplicon Architect. Moreover, the lack of validation of the pipeline using either ground truth datasets or simulation data raises concerns about its accuracy. Additionally, the methodology employed for identifying eccDNA that encompasses multiple gene loci remains unclear.*

We thank Reviewer 2 for pointing out this problem. In the original version of our manuscript, focusing on one eccDNA dataset generated in this study, we have compared the performance between our method and established methods for identification of eccDNA regions, such as Circle_finder, Circle_Map and ecc_finder. Our method has comparable sensitivity and specificity with existing methods, especially Circle_finder and Circle_Map (original Figure 4—figure supplement 2C). We also used one specific genomic region to show that existing methods identified the same eccDNA regions but misassigned the eccDNA boundaries (original Figure 4—figure supplement 2A). In the revised manuscript, we have further included ECCsplorer for comparison. Since Amplicon Architect is more specifically designed for detection of ecDNAs, it was not included in our comparison. Following Reviewer #2' suggestions, we simulated paired-end reads derived from a set of eccDNAs with homologous sequences around breakpoints and employed all methods for eccDNA identification. In total, 97.9%, 97.9%, 97.4%, 95.3% and 91.1% eccDNA regions could be detected by our method, Circle_Map, Circle_finder, ecc_finder and ECCsplorer, respectively (Figure 4—figure supplement 2C). This result suggest that our method has comparable performance in detecting eccDNA regions. However, only our method could faithfully assign breakpoints with 97.4% accuracy, in contrast to no more than 15% by other methods (Figure 4—figure supplement 2D).

As pointed out by Reviewer #2, similar to ECCsplorer, Circle_finder, Circle_Map and ecc_finder, our method fails to identity eccDNAs that encompass multiple gene loci. We have reminded readers of this limitation in our revised manuscript. Besides the schematic workflow (Figure 4—figure supplement 2B), we have included more method details to help readers better understand how our method works (see Methods – eccDNA Detection).

• *Although the author stated that previous studies utilizing short-read sequencing technologies may have incorrectly annotated eccDNA breakpoints, this claim requires careful scrutiny and supporting evidence, which was not provided in the manuscript.*

Following this Reviewer's suggestions, we conducted a systematic evaluation of the performance of various existing methods, namely Circle_finder, Circle_Map, ECCsplorer and ecc_finder, for eccDNA breakpoint annotation.

First, we simulated paired-end reads derived from a set of eccDNAs with homologous sequences around breakpoints and employed all different methods for eccDNA identification. As expected, our method could correctly assign breakpoints for 97.4% eccDNAs (Figure 4—figure supplement 2D), in contrast to no more than 15% by other methods (Figure 4—figure supplement 2D).

Second, we examined the performance of all methods on one dataset generated in this study. Our method detected 59,680, 54,898, 32,993 and 22,019 eccDNAs with homologous sequences that were also detected by Circle_finder, Circle_Map, ECCsplorer and ecc_finder, respectively. Remarkably, we observed that at least 60% of breakpoints were misannotated by the existing methods (Figure 4—figure supplement 2F).

We have included an example in Figure 4—figure supplement 2A, where all existing methods incorrectly annotated the eccDNA breakpoints when homologous sequences were present. These results highlight the advantage of our method over existing methods in accurately annotating eccDNA breakpoints in the presence of homologous sequences.

• *The similarity between the eccDNA profiles of human and mouse sperm remains uncertain, and therefore, analyses of human eccDNA data and comparisons between the two are necessary if the authors claim that their findings of widespread eccDNA formation in mouse spermatogenesis extend to human sperms.*

Our Fig. 5 have shown that human sperm eccDNAs are originated from oligonucleosomal fragmentation (Fig. 5A-C), not associated with meiotic recombination hotspots (Fig. 5D and E) but formed by microhomology directed ligation (Fig. 5F and G). These findings are consistent with what we observed in mouse sperm eccDNAs. To further substantiate our findings, we analyzed an additional eccDNA dataset from human sperms generated by long-read sequencing (Henriksen et al., Mol Cell, 2022). Although predominantly composed of large-sized eccDNAs, the analysis of sequence features also indicated their association with microhomology directed ligation (Figure 5-figure supplement 1E). Overall, the eccDNA profiles in human and mouse sperm exhibit notable similarities.

Reviewer #1 (Recommendations For The Authors):

In the last sentence of the abstract, the authors stated, "provide a potential new way for quality assessment of sperms." There is no basis for the claim in the abstract. The authors need to mention the association of eccDNA with apoptosis somewhere to claim it.

We have revised the Abstract as suggested.

Some of the references need to be clarified. For example, Coquelle et al., 2002 described the BFB cycles and common fragile sites, but the report does not seem to be relevant to eccDNA. Mouakkad-Montoya et al., 2021 enriched eccDNA without rolling-circle amplification.

Thanks for pointing this out. We cited Coquelle et al., 2002 to list known biogenesis mechanisms for ecDNAs but not eccDNAs. We have deleted Mouakkad-Montoya et al., 2021 in our revised manuscript, as it did not involve rolling-circle amplification.

Reviewer #2 (Recommendations For The Authors):

• *It is not clear why the authors took 3000bp as the cutoff to divide eccDNAs into short and long categories. How many long eccDNAs in these samples?*

Henriksen et al identified size range of sperm eccDNAs as ~3–50 kb. We therefore used 3kb as an arbitrary cutoff to better compare two different eccDNA populations with those reported by Henriksen et al. SPA, RST, EST and sperm cells have 278, 609, 373 and 691 eccDNAs respectively that are longer than 3000bp. We have clarified this in the revised manuscript.

• *In figure 2D,2E, what is the zero point in the heatmaps? The 5', 3' end or center of eccDNA? Please make it clear in figure and main text.*

The zero point represents the center of eccDNA regions. We have clarified this in the revised manuscript.

• In line 245, the author mentioned that "periodic distribution of nucleosomes was observed for ~360bp eccDNAs but not for ~180bp ones, indicating that eccDNAs from di-nucleosomes but not mono-nucleosomes preferentially originate from well-positioned nucleosome arrays (Figure 2E)". Please explain how to make the conclusion from the Figure 2E?

Taking the H3K27me3-marked nucleosome as an example, vertical stripes were distributed every ~180bp for ~360bp eccDNAs, as shown by heatmap (more evident if in an enlarged view), and periodic signal distribution was apparent for ~360bp eccDNAs (Figure 2E), as shown by meta-gene analysis on top of heatmap (Figure 2B). However, such pattern was not observed for ~180bp eccDNAs. Similar results could also be observed for nucleosomes marked with other histone variants and histone modifications (H3, H3K27ac, H3K4me1, H3K9ac, H3K36me3, H3K9me3 in Figure 2E). Thus, eccDNAs from di-nucleosomes but not mono-nucleosomes preferentially originate from well-positioned nucleosome arrays in sperm.

• In line 261, the author mentioned: "the large-sized sperm eccDNAs detected in this study also displayed weak but apparent negative correlation with gene density and Alu elements (Figure 3C and D)". However, the data didn't show the "apparent negative correlation", as only one or two data points may support this conclusion and the p-values are not even close to 0.05.

Many thanks for pointing this out. We have toned down this statement as "the large-sized sperm eccDNAs detected in this study displayed a weak negative correlation with gene density or Alu elements (Figure 3C and D)".

• The enrichment of both active (H3K27ac, H3K9ac) and repressive (H3K9me3) histone markers in the original loci of eccDNA poses an intriguing question: how can this seemingly contradictory pattern be explained? In the H3K9me3 heatmap, the average level of H3K9me3 in eccDNA is lower than control's, how to interpret the result?

We found that small-sized eccDNAs were more enriched at H3K27ac-marked euchromatin regions (Figure 2C-E and 3A), while large-sized ones were more enriched at H3K9me3-marked heterochromatin regions (Figure 3A). This is probably because heterochromatin regions are too condensed to be fragmented into smaller pieces for small-sized eccDNA formation, in comparison with euchromatin regions. We have included this information in our revised manuscript.

H3K9me3 histone marks are enriched at repeat sequences that are widely distributed within the mouse genome. Moreover, the H3K9me3 ChIP-seq dataset we analyzed in this study had the highest number of ChIP-seq peaks, compared to ChIP-seq datasets of other histone modifications. Thus, even random control would probably have stronger ChIP-seq signals than small-sized eccDNAs (e.g., ~180bp or ~360bp eccDNAs) that were preferentially generated from active regions.